Università Commerciale Luigi Bocconi

Curie–Weiss & Diffusion

Part II: Reverse OU Dynamics, Collective Closure, and Effective Potentials

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Scope and central question

We study the standard Ornstein–Uhlenbeck diffusion of a high-dimensional clean law $P_0(a)$ and ask when its reverse dynamics can be represented by a low-dimensional potential.
Three logically different statements must be separated:

1. The *score* $\nabla_x \log P_t(x)$ is a gradient by definition.
2. For the standard OU process, the complete reverse drift is also an exact gradient in the full N-dimensional state space.
3. The nontrivial problem is whether this high-dimensional dynamics closes on a small set of collective coordinates. Only after closure do we ask whether the resulting reduced force is generated by a scalar potential.

The Curie–Weiss model provides an exceptionally clean example. Its rank-one interaction, one-dimensional latent magnetization, binary-spin algebra, and large-time branch likelihood combine to close the reverse score on one scalar coordinate (Appendix C.4). We then repeat the calculation for three-state spins, which can exhibit three coexisting magnetization phases without requiring a second collective coordinate at the same leading high-noise order (Appendix B.4). The central two-branch chain is

$$\boxed{\begin{aligned} &P_0(a) \longrightarrow P_t(x) \longrightarrow \nabla_x \log P_t(x) \longrightarrow \text{reverse OU drift} \\ &\qquad \longrightarrow m = \pm m^* \longrightarrow M(x) \longrightarrow \mu(t) \longrightarrow V(\mu, t). \end{aligned}}$$

We also keep the original interaction but replace Ising configurations by the sphere $\|a\|_2 = \sqrt{N}$. This complementary test removes conditional coordinate independence while retaining leading projected-score closure (Appendix D.8).
The three-branch comparison leads to the next question: what is required for a potential with n wells, and when does it close on one scalar rather than several coordinates? The relevant distinction is developed in Appendix B.7.

Notation and approximation hierarchy

In the binary Curie–Weiss sections the clean spin vector is $a = (a_1, \ldots, a_N)$ with $a_i \in \{\pm 1\}$; the three-state extension allows $a_i \in \{-1, 0, +1\}$ (Appendix B.1). The forward diffused vector is $x(t)$, and

$$\Delta_t := T(1 - e^{-2t}).$$

The thermodynamic auxiliary magnetization is m, while

$$M(x) := \frac{1}{N} \sum_{i=1}^{N} x_i, \qquad \mu(t) := \frac{1}{\sqrt{N}} \sum_{i=1}^{N} x_i(t) = \sqrt{N}\, M(x(t)).$$

We use

$$F_i(x, t) := \partial_{x_i} \log P_t(x)$$

for the score, $\zeta_i := x_i e^{-t}/\Delta_t$ for the local noised field, and $Z(\mu, t)$ for the collective field entering the reduced force.
The OU kernel, reverse drift, exact score identity, and Hubbard–Stratonovich transformation are exact. The two-delta-peak thermodynamic reduction uses $N \to \infty$. Closure on $M(x)$ uses the large-time condition $\max_i |\zeta_i| \ll 1$. The simplest displayed potential additionally uses $\Delta_t \simeq T$ (Appendix C.5). The three-branch extension has its own thermodynamic coexistence and high-noise assumptions (Appendix B.2; Appendix B.4).
The spherical extension uses $q(a) = N^{-1} \sum_i a_i$ for the physical clean magnetization and reserves m for the auxiliary HS variable. Its exact two-statistic geometry and high-noise projected closure are separated in Appendix D.7 and Appendix D.8.

Object	Meaning	Status / appendix
$x_t \mid a \sim \mathcal{N}(e^{-t}a, \Delta_t I)$	Forward OU channel	exact (A.1)
$b_{\mathrm{R}} = -b + 2T\nabla \log P_t$	Reverse drift at constant diffusion	exact (A.3)
$F_i = -(x_i - e^{-t}\mathbb{E}[a_i \mid x])/\Delta_t$	OU score identity	exact (A.4)
$P_0(m, a)$: HS augmentation	Latent-variable representation	exact (C.1)
$P_0(m) \Rightarrow W_+\delta_{m^*} + W_-\delta_{-m^*}$	Two-state thermodynamic reduction	large N (C.2)
$\mathbb{E}[a_i \mid x] = \mathcal{G}(M(x), t)$	Scalar closure	large t (C.4)
$V(\mu, t)$ with $\Delta_t = T$	Simple effective potential	large N, large t (C.5)
Three-state $P_0(m)$ and $V_3(\mu, t)$	Three competing magnetization phases	exact $P_0(m)$; asymptotic V_3 (B.2; B.5)
Spherical p_0^q, p_0^{HS}	Physical and auxiliary magnetization laws	exact (D.4)
Spherical $u \cdot F$ and $V_{\mathrm{sph},\Delta}$	Closure of the projected microscopic drift	high-noise (D.8)

Table 1: Exact identities, approximation levels, and clickable appendix derivations.

1 What problem are we actually solving?

Objective. We first isolate the genuinely nontrivial question. This avoids confusing three different uses of the word "potential."

At every time t for which P_t is smooth and strictly positive, the score is

$$F(x, t) = \nabla_x \log P_t(x).$$

It is therefore already a gradient field. This fact alone does not produce a useful low-dimensional description (Appendix A.4).

For the standard OU process, the reverse deterministic drift in an increasing reverse clock is

$$B(x, t) = x + 2T\nabla_x \log P_t(x).$$

Consequently

$$B(x, t) = -\nabla_x U_t(x), \qquad U_t(x) := -\frac{1}{2}\|x\|^2 - 2T \log P_t(x). \tag{1}$$

up to an additive function of time. Thus an exact time-dependent potential already exists in the full N-dimensional space (Appendix A.3).

The research problem is stronger:

$$\boxed{\text{Find a small collection } q(x) \text{ such that the projected reverse dynamics closes on } q.} \tag{2}$$

If the reduced state has one coordinate, a scalar potential can always be obtained locally by integration. For several reduced coordinates, the existence of one scalar potential becomes a nontrivial integrability condition (Appendix A.5).

The central distinction

The score being a gradient is universal. The Curie–Weiss achievement is that the model-specific posterior mean $\mathbb{E}[a_i \mid x]$ becomes a function of one scalar statistic $M(x)$, identical for every coordinate. That closure, not the final integration, is the main analytical result (Appendix C.4; Appendix C.5).

2 Universal OU structure: forward channel, score, and reverse drift

Objective. We derive the part of the argument that depends only on the linear Gaussian OU channel and is therefore available for any clean law P_0.

The forward Itô process is

$$\mathrm{d}x_t = -x_t\,\mathrm{d}t + \sqrt{2T}\,\mathrm{d}W_t, \qquad x(0) = a. \tag{3}$$

Its explicit solution gives

$$\boxed{x(t) \mid a \sim \mathcal{N}(e^{-t}a, \Delta_t I), \qquad \Delta_t = T(1 - e^{-2t}).} \tag{4}$$

Hence

$$P_t(x) = \int \mathrm{d}a\, P_0(a) \frac{1}{(2\pi\Delta_t)^{N/2}} \exp\left[-\frac{\|x - e^{-t}a\|^2}{2\Delta_t}\right]. \tag{5}$$

The OU kernel and convolution follow directly from the explicit solution (Appendix A.1).

Differentiating this convolution gives the exact score identity

$$\boxed{F_i(x,t) = \partial_{x_i} \log P_t(x) = -\frac{x_i - e^{-t}\mathbb{E}[a_i \mid x]}{\Delta_t}.} \tag{6}$$

All dependence on the clean law is concentrated in the denoiser $\mathbb{E}[a_i \mid x]$ (Appendix A.4).

To state the reverse dynamics without a sign ambiguity, introduce an increasing reverse clock

$$s := t_f - t.$$

Then the reverse process is

$$\boxed{\mathrm{d}y_s = \left[y_s + 2T\nabla_y \log P_{t_f - s}(y_s)\right]\,\mathrm{d}s + \sqrt{2T}\,\mathrm{d}\overline{W}_s.} \tag{7}$$

In the decreasing forward-time convention this is

$$\boxed{-\frac{\mathrm{d}y_i}{\mathrm{d}t} = y_i + 2TF_i(y,t) + \eta_i(t).} \tag{8}$$

The connection from Itô calculus to the Fokker–Planck equation is derived in Appendix A.2; the time-reversal argument and its factor $2T$ are derived in Appendix A.3.

3 Curie–Weiss structure and the emergence of scalar closure

Objective. We now identify the special structural ingredients that make the generic OU score in (6) close on a single collective coordinate.

The clean law is

$$P_0(a) \propto \exp\left[\frac{\beta}{2N}\left(\sum_{i=1}^N a_i\right)^2 + \frac{h}{N}\sum_{i=1}^N a_i\right], \qquad a_i \in \{\pm 1\}. \tag{9}$$

The interaction depends only on the rank-one statistic $\sum_i a_i$. A one-dimensional Hubbard–Stratonovich variable therefore linearizes the quadratic term, yielding

$$P_0(m,a) \propto \exp\left(-\frac{\beta N}{2}m^2\right) \prod_{i=1}^N \exp[(\beta m + h/N)a_i]\,. \tag{10}$$

Conditional on m, the spins are independent (Appendix C.1). Summing them out gives

$$P_0(m) \asymp e^{-\beta N \Phi_N(m)}, \qquad \Phi_N(m) = \frac{m^2}{2} - \frac{1}{\beta}\log\cosh(\beta m + h/N). \tag{11}$$

For $\beta > 1$, Laplace concentration produces two stable branches near $\pm m^*$, where

$$m^* = \tanh(\beta m^*). \tag{12}$$

Their normalized masses are

$$W_\pm = \frac{e^{\pm hm^*}}{2\cosh(hm^*)}. \tag{13}$$

Thus, at leading order in N, the thermodynamic latent variable becomes a two-state label (Appendix C.2).

Conditional on branch $\pm$, the OU-diffused branch likelihood contains the branch-dependent factor

$$Q_\pm(x) = \prod_{j=1}^{N} [1 \pm m^* \tanh \zeta_j], \qquad \zeta_j := \frac{x_j e^{-t}}{\Delta_t}. \tag{14}$$

The exact branch posterior within the two-state approximation is

$$\boxed{P(\pm \mid x) = \frac{W_\pm Q_\pm(x)}{W_+ Q_+(x) + W_- Q_-(x)}.} \tag{15}$$

This formula displays the prior branch mass and the diffused branch likelihood separately (Appendix C.3).

The cleanest large-time reduction is obtained from the likelihood ratio, not from the two absolute likelihoods separately:

$$\begin{aligned} \log \frac{Q_+(x)}{Q_-(x)} &= \sum_{j=1}^{N} \log \frac{1 + m^* \tanh \zeta_j}{1 - m^* \tanh \zeta_j} \\ &= 2 \sum_{j=1}^{N} \operatorname{artanh}(m^* \tanh \zeta_j) \\ &= 2m^* \sum_{j=1}^{N} \zeta_j + O\left(\sum_{j=1}^{N} |\zeta_j|^3 \right). \end{aligned} \tag{16}$$

There is no quadratic term in the ratio; the sign-independent quadratic contribution cancels exactly. Since

$$\sum_j \zeta_j = \frac{e^{-t}}{\Delta_t} \sum_j x_j = NM(x) \frac{e^{-t}}{\Delta_t},$$

the branch posterior depends, at leading order, only on $M(x)$ (Appendix C.4).

At the same order, the branch-conditional local means become $\pm m^*$. Combining them with (15) gives

$$\boxed{\mathbb{E}[a_i \mid x] \simeq m^* \tanh\left[m^* \left(h + NM(x) \frac{e^{-t}}{\Delta_t} \right) \right].} \tag{17}$$

The right-hand side is independent of the site label i. Substitution into (6) gives the closed large-time score

$$\boxed{F_i^{\mathrm{LT}}(x,t) = -\frac{x_i}{\Delta_t} + m^* \frac{e^{-t}}{\Delta_t} \tanh\left[m^* \left(h + NM(x) \frac{e^{-t}}{\Delta_t} \right) \right].} \tag{18}$$

The peak masses come from Appendix C.2, the exact branch algebra from Appendix C.3, and the controlled high-noise approximation from Appendix C.4.

Why the dimensional reduction works

The interaction is rank one, so one auxiliary magnetization is sufficient. Permutation symmetry makes every coordinate couple to that same latent variable. At large time, the branch log-likelihood ratio accumulates the weak local evidence only through $\sum_i x_i$. These three facts turn the model-specific denoiser into the same scalar function of $M(x)$ at every site (Appendix C.1; Appendix C.4).

4 Closed collective dynamics and the effective potential

Objective. We project the closed score onto the collective coordinate and distinguish the existence of a potential from the additional approximation $\Delta_t \simeq T$.

Define

$$\boxed{\mu(t) := \frac{1}{\sqrt{N}} \sum_{i=1}^{N} x_i(t) = \sqrt{N}\, M(x(t)).} \tag{19}$$

The projected noise

$$\eta_\mu(t) := \frac{1}{\sqrt{N}} \sum_i \eta_i(t)$$

satisfies

$$\langle \eta_\mu(t)\eta_\mu(t')\rangle = 2T\delta(t-t'). \tag{20}$$

The noise calculation is given in Appendix C.5; its linear projection mechanism is also recorded in Appendix A.5. Summing the reverse equations and inserting (18) gives, before approximating Δ_t,

$$-\frac{\mathrm{d}\mu}{\mathrm{d}t} = \left(1 - \frac{2T}{\Delta_t}\right)\mu + 2Tm^*\sqrt{N}\frac{e^{-t}}{\Delta_t}\tanh\left[m^*\left(h + \sqrt{N}\mu\frac{e^{-t}}{\Delta_t}\right)\right] + \eta_\mu(t). \tag{21}$$

This one-dimensional force already has a potential. Define

$$\boxed{V_\Delta(\mu,t) = \frac{1}{2}\left(\frac{2T}{\Delta_t} - 1\right)\mu^2 - 2T\log\cosh\left[m^*\left(h + \frac{\sqrt{N}e^{-t}}{\Delta_t}\mu\right)\right].} \tag{22}$$

Then

$$-\frac{\mathrm{d}\mu}{\mathrm{d}t} = -\partial_\mu V_\Delta(\mu,t) + \eta_\mu(t). \tag{23}$$

Thus $\Delta_t \simeq T$ is not needed for the existence of a potential. It is only needed for the simpler form used near the high-noise transition (Appendix C.5).

At large forward time,

$$\Delta_t = T(1 - e^{-2t}) \simeq T,$$

so

$$\boxed{-\frac{\mathrm{d}\mu}{\mathrm{d}t} = -\mu + 2m^*\sqrt{N}e^{-t}\tanh Z(\mu,t) + \eta_\mu(t),} \tag{24}$$

where

$$Z(\mu,t) := m^*\left(h + \frac{\sqrt{N}}{T}e^{-t}\mu\right). \tag{25}$$

Using $(\log\cosh Z)' = \tanh Z$, the simplified potential is

$$\boxed{V(\mu,t) = \frac{1}{2}\mu^2 - 2T\log\cosh Z(\mu,t).} \tag{26}$$

The reduction to this expression is verified by differentiation in Appendix C.5.

For stability it is preferable to return to the increasing reverse clock $s = t_f - t$. The deterministic reduced equation is then

$$\frac{\mathrm{d}\mu}{\mathrm{d}s} = -\partial_\mu V(\mu, t_f - s). \tag{27}$$

Therefore a stationary point is locally stable precisely when it is a local minimum of V, equivalently when $V'' > 0$ (Appendix C.5).

5 How the potential changes during reverse diffusion

Objective. We interpret the time-dependent potential and locate the loss of stability of the uninformative state.

In the symmetric case $h = 0$, the origin is stationary. Expanding $\log\cosh Z = Z^2/2 + O(Z^4)$ gives

$$V(\mu, t) = \frac{1}{2}\left[1 - \frac{2m^{*2}N}{T}e^{-2t}\right]\mu^2 + O(\mu^4). \tag{28}$$

Hence

$$V''(0, t) = 1 - \frac{2m^{*2}N}{T}e^{-2t}. \tag{29}$$

The origin loses stability when

$$e^{-2t} = \frac{T}{2m^{*2}N}, \qquad \text{so} \qquad \boxed{t_{\text{spec}} = \frac{1}{2}\log\left(\frac{2m^{*2}N}{T}\right).} \tag{30}$$

These curvature and speciation expressions are evaluated in Appendix C.5. In particular,

$$t_{\text{spec}} = \frac{1}{2}\log N + O(1).$$

Later in reverse generation, t is smaller and the collective argument $|Z|$ becomes large. Since

$$\log\cosh Z = |Z| - \log 2 + o(1),$$

the symmetric potential becomes, up to μ-independent terms,

$$V(\mu, t) \simeq \frac{1}{2}\mu^2 - 2m^*\sqrt{N}e^{-t}|\mu|. \tag{31}$$

The origin has become unstable and two ferromagnetic wells have formed (Appendix C.5).

The local expansion is self-consistent at the transition. For typical high-noise coordinates,

$$\max_i |x_i| = O_{\mathbb{P}}(\sqrt{T\log N}),$$

while $e^{-t_{\text{spec}}} = O(N^{-1/2})$. Thus

$$\max_i |\zeta_i| = O_{\mathbb{P}}\left(\sqrt{\frac{\log N}{N}}\right) \longrightarrow 0. \tag{32}$$

At the same time,

$$\Delta_{t_{\text{spec}}} = T\left(1 - O(N^{-1})\right),$$

which justifies $\Delta_t \simeq T$ at the transition scale (Appendix C.4; Appendix C.5).

6 From two wells to three: a first structural test

Objective. We summarize the three-magnetization extension before asking what changes for an arbitrary number of wells. The full calculation follows the same steps as the binary model in Appendix B.

Allow $a_i \in \{-1, 0, +1\}$ and add an occupancy penalty to the Curie–Weiss interaction:

$$P_0(a) \propto \exp\left[\frac{\beta}{2N}\left(\sum_i a_i\right)^2 - \lambda\sum_i a_i^2\right]. \tag{33}$$

The same one-scalar Hubbard–Stratonovich step gives an exact marginal $P_0(m) \propto e^{-\beta N\Phi_\lambda(m)}$, with

$$\Phi_\lambda(m) = \frac{m^2}{2} - \frac{1}{\beta}\log\left[1 + 2e^{-\lambda}\cosh(\beta m)\right]. \tag{34}$$

Unlike the Ising case, suitable coexistence parameters produce three thermodynamic minima $-m^*$, 0, and $+m^*$. For instance, at $\beta = 4$ one finds $\lambda \simeq 1.90506$ and $m^* \simeq 0.73148$. Equal free-energy depths ensure that all three phases survive at large N, but their masses need not be $1/3$: the Laplace curvatures determine the weights $W_- = W_+ = W$ and W_0 (Appendix B.1; Appendix B.2).

Passing the three branches through the same OU channel gives an exact one-dimensional integral for $P_t(m \mid x)$, which still depends on the individual x_i (Appendix B.3). At the high-noise scale $t = \frac{1}{2}\log N + O(1)$, however, the branch evidence depends to leading order only on $\mu = N^{-1/2}\sum_i x_i$; the different branch variances cancel from the leading likelihood ratio after concentration (Appendix B.4). With $c_t = m^*\sqrt{N}e^{-t}$ and $\Delta_t \simeq T$, the resulting candidate scalar reverse potential is

$$\boxed{V_3(\mu, t) \simeq \frac{1}{2}\mu^2 - 2T\log\Big[W_0 + 2We^{-c_t^2/(2T)}\cosh\Big(\frac{c_t\mu}{T}\Big)\Big] + C(t).} \tag{35}$$

The central branch contributes an extra term inside the logarithm; three clean phases do not guarantee three visible wells at every reverse time. This is a leading-order *full-score closure* result, distinct from the exact potential of the projected one-dimensional OU marginal (Appendix B.5; Appendix B.6).

Emerging trend. The two-branch and three-branch models both close on the *same one* collective coordinate at the stated high-noise order. Hence the number of thermodynamic phases does not, by itself, fix the number of latent coordinates. The question for n wells is whether branch evidence and within-branch denoising can still be represented by a small set of collective statistics (Appendix B.7).

7 Relaxing the Ising constraint: spherical Curie–Weiss

Objective. The three-state example changes the number of phases while retaining a conditional product structure. We now make a different test: keep the original quadratic interaction and its two-phase regime, but replace binary coordinates by a continuous vector with *only a global norm constraint*. The calculation in Appendix D asks whether conditional independence is essential for closure, or whether rotational symmetry and concentration can replace it.

We take uniform probability measure σ_N on the sphere $\{a \in \mathbb{R}^N : \|a\|_2 = \sqrt{N}\}$ and define

$$\boxed{P_0^{\mathrm{sph}}(\,\mathrm{d}a) = \frac{1}{Z_N^{\mathrm{sph}}}\exp\left[\frac{\beta}{2N}\Big(\sum_i a_i\Big)^2 + \frac{h}{N}\sum_i a_i\right]\,\mathrm{d}\sigma_N(a).} \tag{36}$$

Here Z_N^{sph} is the partition function, $\beta > 0$ is the interaction parameter, and h is a fixed weak-field parameter. The measure $\mathrm{d}\sigma_N(a)$ is not an additional energy term: it specifies how continuous configurations are counted. Both Ising and spherical configurations have squared norm N. For a macroscopic physical magnetization $q(a) := N^{-1}\sum_i a_i$, the exponent is $\beta N q^2/2 + hq$: the interaction remains extensive, of order N, while the field bias remains of order one (Appendix D.1).

The Hubbard–Stratonovich identity is unchanged, but the integral over the sphere does not factor into single-site partition functions. With $u = N^{-1/2}(1, \dots, 1)$, write $a = \sqrt{N}[qu + \sqrt{1-q^2}\,\omega]$, where ω is a unit vector orthogonal to u. The exact surface change of variables is

$$\mathrm{d}\sigma_N(a) = C_N(1-q^2)^{(N-3)/2}\,\mathrm{d}q\,\mathrm{d}\bar{\sigma}_{N-2}(\omega), \qquad C_N = \frac{\Gamma(N/2)}{\sqrt{\pi}\,\Gamma((N-1)/2)}. \tag{37}$$

The angular measure $\bar{\sigma}_{N-2}$ is normalized on the transverse unit sphere. The factor multiplying $\mathrm{d}q$ is derived from the surface Jacobian, including the substitution $q = \cos\theta$, in Appendix D.3. Integrating the angular variable gives the *physical* magnetization density

$$\boxed{p_0^q(q) = \frac{C_N}{Z_N^{\mathrm{sph}}}(1-q^2)^{(N-3)/2}e^{\beta N q^2/2 + hq}, \qquad -1 < q < 1.} \tag{38}$$

The auxiliary Hubbard–Stratonovich variable $m \in \mathbb{R}$ is different: exactly, $m \mid a \sim \mathcal{N}(q(a), 1/(\beta N))$, so its density is p_0^q smoothed by that Gaussian. Both laws, with every normalization constant, are obtained in Appendix D.2 and Appendix D.4.

For fixed $\beta > 1$, Laplace concentration yields two physical phases at

$$\boxed{m^*_{\mathrm{sph}} = \sqrt{1-\beta^{-1}}, \qquad q \simeq \pm m^*_{\mathrm{sph}}, \qquad W_\pm = \frac{e^{\pm h m^*_{\mathrm{sph}}}}{2\cosh(h m^*_{\mathrm{sph}})}.} \tag{39}$$

Thus the microscopic geometry changes the spontaneous magnetization from $m^* = \tanh(\beta m^*)$ to the explicit spherical expression, but preserves two symmetry-related phases. This result is derived both from the physical magnetization density and from the joint Hubbard–Stratonovich saddle in Appendix D.5.

After the same isotropic OU channel, rotational symmetry gives an exact ambient density $P_t(x) = G_t(\mu, r)$, where $\mu = u \cdot x$ and $r = \|x - \mu u\|$. The projected full score need not be a function of μ alone at finite N. Its exact identity is

$$\boxed{u \cdot F(x,t) = -\frac{\mu}{\Delta_t} + \frac{\sqrt{N}e^{-t}}{\Delta_t}\mathbb{E}[q(a) \mid x].} \tag{40}$$

The spherical likelihood and both latent posteriors are evaluated in Appendix D.6; the two-statistic score and its interpretation are spelled out in Appendix D.7.

For $t_N = \frac{1}{2}\log N + \tau$ with bounded τ, fixed $T > 0$, fixed $\beta > 1$, and fixed h, the transverse radius concentrates and its angular likelihood is controlled uniformly in q. The resulting posterior mean satisfies

$$\mathbb{E}[q(a) \mid X_{t_N}] = m^*_{\mathrm{sph}} \tanh\left[m^*_{\mathrm{sph}}\left(h + \frac{\sqrt{N}e^{-t_N}}{\Delta_{t_N}}\mu\right)\right] + o_{\mathbb{P}}(1). \tag{41}$$

Here $o_{\mathbb{P}}(1)$ means a remainder tending to zero in probability under the noised spherical law as $N \to \infty$. The proof in Appendix D.8 uses the exact posterior rather than differentiating an uncontrolled likelihood approximation. This establishes closure of the *projection of the full microscopic reverse drift*, at the stated high-noise scale, without assuming independent clean coordinates.

Integration of that leading scalar drift gives

$$\boxed{\begin{aligned} V_{\mathrm{sph},\Delta}(\mu,t) = {} & \frac{1}{2}\left(\frac{2T}{\Delta_t} - 1\right)\mu^2 \\ & - 2T \log\cosh\left[m^*_{\mathrm{sph}}\left(h + \frac{\sqrt{N}e^{-t}}{\Delta_t}\mu\right)\right] + C(t). \end{aligned}} \tag{42}$$

The arbitrary function $C(t)$ does not affect the force. Setting $\Delta_t \simeq T$ recovers the same functional form as the Ising potential, with the spherical value of m^* (Appendix D.9). This is not an exact finite-N scalar closure and is not claimed down to $t = 0$. The exact one-dimensional *marginal* reverse potential is treated separately in Appendix D.10.

Structural lesson. Increasing the phase count and changing the microscopic geometry are distinct tests. The spherical calculation keeps two phases but removes conditional coordinate independence. Its leading projected potential survives because only the uniform magnetization carries unresolved phase information, while the transverse contribution cancels between idealized symmetric branches or concentrates in the full model. The comparison in Appendix D.11 therefore motivates studying both the number of phases and the geometry of the remaining informative observables in the n-phase problem below.

8 From three wells to n wells: the next closure problem

Objective. We now state the next research problem: how does the smallest closing representation change when the number of phases grows, possibly with several independent directions? The Gaussian branch-geometry example in Appendix B.7 clarifies why the number of wells and the latent dimension need not coincide. The projection and potential conditions are spelled out in Appendix A.5.

Let

$$h^1, \ldots, h^p \in \mathbb{R}^D, \qquad \|h^\alpha\|_2 = 1,$$

with no initial assumption on their overlaps. A genuine latent-template model is

$$P_0(\alpha) = w_\alpha, \qquad P_0(a \mid \alpha) = \frac{1}{Z_\alpha} \exp(\beta \langle a, h^\alpha \rangle)\,, \tag{43}$$

so that

$$\boxed{P_0(a) = \sum_{\alpha=1}^{p} w_\alpha P_0(a \mid \alpha) \propto \sum_{\alpha=1}^{p} \widetilde{w}_\alpha e^{\beta \langle a, h^\alpha \rangle}.} \tag{44}$$

This must be distinguished from

$$\exp\left(\beta \sum_{\alpha=1}^{p} \langle a, h^\alpha \rangle\right) = \exp\left(\beta \left\langle a, \sum_{\alpha} h^\alpha \right\rangle\right),$$

which is only one effective linear field and does not represent competing latent templates (Appendix B.7).

The universal OU identities remain unchanged:

$$F_i(x,t) = -\frac{x_i - e^{-t}\mathbb{E}[a_i \mid x]}{\Delta_t}, \qquad b_{\mathrm{R}} = x + 2T\nabla \log P_t.$$

The model-specific question is whether $\mathbb{E}[a \mid x]$ depends on x only through finitely many overlaps, naturally (Appendix B.7)

$$q_\alpha(x) := \langle h^\alpha, x \rangle, \qquad \alpha = 1, \dots, p. \tag{45}$$

A useful closure would have the form

$$\mathbb{E}[a \mid x] = \mathcal{D}(q_1(x), \dots, q_p(x), t) + \text{controlled remainder}. \tag{46}$$

Projecting isotropic OU noise onto these coordinates gives correlated noise:

$$\langle \eta_\alpha(t) \eta_\beta(t') \rangle = 2T G_{\alpha\beta} \delta(t - t'), \qquad G_{\alpha\beta} := \langle h^\alpha, h^\beta \rangle. \tag{47}$$

Thus the Gram matrix is unavoidable even before any additional structure is assumed (Appendix A.5).

If the closed reduced drift is $\mathcal{F}(q,t)$, an ordinary Euclidean potential

$$\mathcal{F} = -\nabla_q V$$

requires, on a simply connected domain,

$$\boxed{\partial_{q_\beta} \mathcal{F}_\alpha = \partial_{q_\alpha} \mathcal{F}_\beta \qquad \text{for all } \alpha, \beta.} \tag{48}$$

If the natural mobility is the Gram matrix and $\mathcal{F} = -G\nabla_q V$, then the corresponding condition is that $G^{-1}\mathcal{F}$ be curl free on the nondegenerate template span (Appendix A.5).

Research program suggested by the Curie–Weiss example

The next investigation should proceed in this order:

1. derive the exact posterior/score for the p-template mixture;
2. identify the smallest set of overlaps on which it closes, exactly or asymptotically (Appendix B.7);
3. derive the projected noise covariance and drift, retaining the Gram geometry (Appendix A.5);
4. test whether the reduced force is conservative in the appropriate metric (Appendix A.5);
5. only then study the shape, bifurcations, and possible hierarchical wells of the resulting potential.

9 Main conclusions

Objective. We connect the two- and three-well results and formulate the problem that remains for an arbitrary number of wells and directions.

The full OU reverse drift has an exact potential in the original state space, and the score is always a gradient (Appendix A.3; Appendix A.4). Neither fact explains the special Curie–Weiss reduction: in the symmetric binary model, two macroscopic branches lead to a denoiser depending on the mean $M(x)$ alone, and therefore to a one-dimensional double-well potential (Appendix C.2; Appendix C.4; Appendix C.5).

Changing the clean law to the three-state model yields three coexisting magnetization phases but preserves scalar closure at the same leading high-noise order, giving the candidate potential (35) (Appendix B.2; Appendix B.4; Appendix B.5). This is the trend motivating our next question: for n wells, which assumptions keep the relevant information one-dimensional, and which demand several independent collective coordinates? The difference between branch sufficiency and actual reverse-score closure is explained in Appendix B.7; the additional metric and integrability issues for multi-directional models are derived in Appendix A.5.

The spherical comparison changes the reference measure instead of the phase count. Its exact physical and auxiliary magnetization laws differ at finite N, while both concentrate on the same two thermodynamic branches (Appendix D.4; Appendix D.5). Rotation symmetry reduces the noisy density to the longitudinal coordinate and a transverse radius. At the high-noise scale, the projected full-state drift closes on μ and gives the same functional potential with a different spontaneous magnetization (Appendix D.8; Appendix D.9). This shows, within the stated regime, that conditional independence is not necessary for scalar projected closure; it does not remove the need to control hidden observables in other models or at lower noise.

Appendices

Appendix A records the universal OU derivation concisely. Appendix B follows the same reasoning for three-state spins and compares exact results with their high-noise closure. Appendix C retains the continuous, detailed Ising Curie–Weiss derivation from the clean prior to the potential. Appendix D adds the spherical comparison, including its reference measure, surface Jacobian, exact magnetization laws, and controlled projected-score closure. The three preceding appendices are retained.

A Universal OU machinery: concise derivation

Objective. This appendix supplies the universal stochastic-calculus steps used in the main text without interrupting the Curie–Weiss argument.

A.1 Forward solution and Gaussian channel

For one coordinate,

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} + x_i = \eta_i(t), \qquad x_i(0) = a_i.$$

Multiplication by e^t and integration give

$$x_i(t) = e^{-t}a_i + \int_0^t e^{-(t-s)}\eta_i(s)\,\mathrm{d}s.$$

With

$$\langle \eta_i(s)\eta_j(s')\rangle = 2T\delta_{ij}\delta(s-s'),$$

the accumulated-noise variance is

$$2T\int_0^t e^{-2(t-s)}\,\mathrm{d}s = T(1-e^{-2t}) = \Delta_t.$$

This proves (4) and (5).

A.2 Itô generator and Fokker–Planck equation

For

$$\mathrm{d}X_i = b_i(X,t)\,\mathrm{d}t + \sum_k \sigma_{ik}(X,t)\,\mathrm{d}W_k,$$

Itô's formula gives the generator

$$\mathcal{L}f = \sum_i b_i\partial_i f + \frac{1}{2}\sum_{i,j} a_{ij}\partial_i\partial_j f, \qquad a = \sigma\sigma^\top.$$

Transferring derivatives from a test function to the density gives

$$\partial_t P = -\sum_i \partial_i(b_i P) + \frac{1}{2}\sum_{i,j}\partial_i\partial_j(a_{ij}P). \tag{49}$$

For $b = -x$ and $a = 2TI$,

$$\partial_t P_t = \nabla\cdot(xP_t) + T\nabla^2 P_t. \tag{50}$$

A.3 Probability current and reverse drift

For constant isotropic diffusion,

$$\partial_t P = -\nabla \cdot J, \qquad J = bP - T\nabla P.$$

Define $s = t_f - t$ and $Q(y, s) = P_{t_f - s}(y)$. The required reversed density evolution is

$$\partial_s Q = -\nabla \cdot (yQ) - T\nabla^2 Q.$$

A reverse SDE with drift B has

$$\partial_s Q = -\nabla \cdot (BQ) + T\nabla^2 Q.$$

Matching the local currents, with the usual decay/boundary conditions and no additional solenoidal current, gives

$$BQ = yQ + 2T\nabla Q,$$

hence

$$B = y + 2T\nabla \log Q.$$

This proves (7) and (8). The factor $2T$ appears because the diffusive current $-T\nabla P$ must become $+T\nabla P$ when the probability flow is reversed. In particular, the full-space OU reverse drift has the exact gradient representation

$$-\nabla_x \left[-\frac{1}{2}\|x\|^2 - 2T \log P_t(x) \right] = x + 2T\nabla_x \log P_t(x).$$

A.4 Exact OU score identity

Differentiate (5):

$$\partial_{x_i} P_t(x) = -\frac{x_i}{\Delta_t} P_t(x) + \frac{e^{-t}}{\Delta_t} \int \mathrm{d}a \, a_i P_t(a, x). \tag{51}$$

After division by $P_t(x)$,

$$\partial_{x_i} \log P_t(x) = -\frac{x_i}{\Delta_t} + \frac{e^{-t}}{\Delta_t} \int \mathrm{d}a \, a_i P_t(a \mid x),$$

which is (6).

A.5 Linear projections, noise geometry, and the potential criterion

For a matrix $H \in \mathbb{R}^{p \times D}$, define $q = Hx$ for a D-dimensional OU process $\mathrm{d}x = -x \, \mathrm{d}t + \sqrt{2T} \, \mathrm{d}W$. Linearity of the Itô differential gives

$$\mathrm{d}q = -q \, \mathrm{d}t + \sqrt{2T} \, H \, \mathrm{d}W, \qquad \mathrm{d}\langle q_\alpha, q_\beta \rangle_t = 2T(HH^\top)_{\alpha\beta} \, \mathrm{d}t.$$

If the rows of H are the template vectors $(h^\alpha)^\top$, then $G = HH^\top$ is their Gram matrix, independently of whether these directions are orthogonal. For any smooth scalar $V(q)$, the chain rule yields

$$H\nabla_x [V(Hx)] = HH^\top \nabla_q V = G\nabla_q V.$$

Thus projection of an ambient gradient component that factors through q naturally produces a *metric gradient* $-G\nabla_q V$, not necessarily a Euclidean gradient $-\nabla_q V$. This statement presupposes that the rest of the ambient drift also closes on q. For a smooth closed force $\mathcal{F}(q)$ on a simply connected nondegenerate domain, a Euclidean potential exists if and only if $\partial_{q_\beta} \mathcal{F}_\alpha = \partial_{q_\alpha} \mathcal{F}_\beta$. If G is constant and invertible, a metric potential instead exists if and only if $G^{-1}\mathcal{F}$ has symmetric cross-derivatives. For singular G, the statement must be restricted to the independent template span.

B Three magnetization setting: looking for a trend

Objective. We repeat the sequence of the binary Curie–Weiss derivation, but allow $a_i \in \{-1, 0, +1\}$. The central questions are whether the clean measure can support three competing magnetizations, whether their OU-diffused likelihoods can be evaluated, and whether the reverse force still depends only on $\mu = N^{-1/2}\sum_i x_i$. We will distinguish exact finite-N identities from the three-branch and high-noise approximations. The original two-magnetization calculation is developed in full in Appendix C.

B.1 Clean prior and the Hubbard–Stratonovich variable

We begin with the mean-field three-state (Blume–Capel-type) law

$$\boxed{P_0(a) = \frac{1}{\mathcal{Z}_N}\exp\left[\frac{\beta}{2N}\left(\sum_{i=1}^N a_i\right)^2 + \frac{h}{N}\sum_{i=1}^N a_i - \lambda\sum_{i=1}^N a_i^2\right], \qquad a_i \in \{-1, 0, +1\}.} \tag{52}$$

The quadratic interaction is identical to Curie–Weiss. The new parameter λ controls the occupancy of the nonzero states, since $a_i^2 = 1$ for $a_i = \pm 1$ and $a_i^2 = 0$ for $a_i = 0$. Set

$$S(a) := \sum_i a_i, \qquad Q(a) := \sum_i a_i^2.$$

Here S records the magnetization while Q counts occupied (± 1) sites. The new term does not remove the rank-one interaction; thus one scalar Hubbard–Stratonovich variable still suffices. Using the exact Gaussian identity (92) with $S = S(a)$,

$$e^{\beta S^2/(2N)} = \sqrt{\frac{\beta N}{2\pi}}\int_{\mathbb{R}} \mathrm{d}m\, \exp\left[-\frac{\beta N}{2}m^2 + \beta m S\right]. \tag{53}$$

Introduce the shorthand $u_m := \beta m + h/N$. Inserting the identity into (52) gives the augmented joint law

$$P_0(m, a) = \frac{1}{\widetilde{\mathcal{Z}}_N} e^{-\beta N m^2/2}\prod_{i=1}^N e^{u_m a_i - \lambda a_i^2}. \tag{54}$$

Integrating out m returns the original distribution of a. Conditional on m the coordinates become independent, with

$$\boxed{P_0(a \mid m) = \prod_{i=1}^N \frac{e^{u_m a_i - \lambda a_i^2}}{D_\lambda(u_m)}, \qquad D_\lambda(u) := 1 + 2e^{-\lambda}\cosh u.} \tag{55}$$

Indeed, the one-site normalization is the elementary sum

$$\sum_{a_i=-1,0,+1} e^{u a_i - \lambda a_i^2} = e^{-u-\lambda} + 1 + e^{u-\lambda} = 1 + 2e^{-\lambda}\cosh u. \tag{56}$$

It follows, still exactly at finite N, that

$$\begin{aligned} P_0(a_i = 0 \mid m) &= \frac{1}{D_\lambda(u_m)}, \\ P_0(a_i = \pm 1 \mid m) &= \frac{e^{-\lambda \pm u_m}}{D_\lambda(u_m)}. \end{aligned} \tag{57}$$

This replaces the Ising single-site partition function $2\cosh u$ by $1 + 2e^{-\lambda}\cosh u$. No other change was required to perform the decoupling.

B.2 Exact $P_0(m)$, coexistence, and three branch weights

Now sum (54) over a. Because the sites factorize,

$$\begin{aligned} P_0(m) &\propto e^{-\beta N m^2/2} \prod_{i=1}^{N} \sum_{a_i=-1,0,+1} e^{u_m a_i - \lambda a_i^2} \\ &= e^{-\beta N m^2/2} [D_\lambda(u_m)]^N. \end{aligned} \tag{58}$$

Consequently the normalized marginal of the *continuous auxiliary* variable m is

$$\boxed{P_0(m) = \frac{e^{-\beta N m^2/2}[1 + 2e^{-\lambda}\cosh(\beta m + h/N)]^N}{\displaystyle\int_{\mathbb{R}} \mathrm{d}v\, e^{-\beta N v^2/2}[1 + 2e^{-\lambda}\cosh(\beta v + h/N)]^N}.} \tag{59}$$

As before, write $P_0(m) \propto e^{-\beta N \Phi_{\lambda,N}(m)}$, with

$$\boxed{\Phi_{\lambda,N}(m) = \frac{m^2}{2} - \frac{1}{\beta}\log[1 + 2e^{-\lambda}\cosh(\beta m + h/N)].} \tag{60}$$

Thus $P_0(m)$ is known as an exact one-dimensional expression, even before any saddle-point approximation.

The derivative is

$$\Phi'_{\lambda,N}(m) = m - \frac{2e^{-\lambda}\sinh(\beta m + h/N)}{1 + 2e^{-\lambda}\cosh(\beta m + h/N)}. \tag{61}$$

At $h = 0$, a stationary magnetization m_α therefore solves

$$\boxed{m_\alpha = \frac{2e^{-\lambda}\sinh(\beta m_\alpha)}{1 + 2e^{-\lambda}\cosh(\beta m_\alpha)}.} \tag{62}$$

The right-hand side is also $\mathbb{E}[a_i \mid m_\alpha]$ by (57), so a stationary m equals the conditional mean spin. The central solution $m_0 = 0$ always exists; for appropriate (β, λ) there are additional stable solutions $m_\pm = \pm m^*$.

To have three *coexisting macroscopic phases*, mere stationarity is insufficient: all three points must be local minima and have the same leading free-energy value. At $h = 0$, symmetry equates the outer two, so the coexistence conditions reduce to

$$\begin{cases} m^* = \dfrac{2e^{-\lambda}\sinh(\beta m^*)}{1 + 2e^{-\lambda}\cosh(\beta m^*)}, \\ \dfrac{\beta (m^*)^2}{2} = \log\left[\dfrac{1 + 2e^{-\lambda}\cosh(\beta m^*)}{1 + 2e^{-\lambda}}\right], \end{cases} \tag{63}$$

together with $\Phi''_\lambda(0) > 0$ and $\Phi''_\lambda(m^*) > 0$. The second equation is just $\Phi_\lambda(m^*) = \Phi_\lambda(0)$ written out explicitly. For example, solving (63) at $\beta = 4$ gives

$$\lambda_{\mathrm{coex}} \simeq 1.90506335, \qquad m^* \simeq 0.73148205.$$

This is a concrete three-phase setting, rather than an arbitrary three-component mixture imposed by hand.

To calculate the masses of the peaks, expand Φ quadratically around each minimum. Laplace's method gives

$$I_\alpha \simeq e^{-\beta N \Phi(m_\alpha)} \sqrt{\frac{2\pi}{\beta N \Phi''(m_\alpha)}}, \qquad \alpha \in \{-, 0, +\}. \tag{64}$$

At coexistence the exponential factors agree, but the curvatures need not agree. Hence, for $h = 0$,

$$\boxed{W_\alpha = \frac{[\Phi''(m_\alpha)]^{-1/2}}{\displaystyle\sum_{\gamma\in\{-,0,+\}} [\Phi''(m_\gamma)]^{-1/2}} + o(1), \quad W_- = W_+.} \tag{65}$$

For $\beta = 4$ at coexistence, the weights are approximately

$$W_- \simeq 0.282, \qquad W_0 \simeq 0.436, \qquad W_+ \simeq 0.282.$$

Equal free-energy depths do *not* imply three exactly equal masses. Unlike the two Ising branches, the middle peak is not the mirror image of an outer peak. A finite-size adjustment of λ of order N^{-1} can balance the asymptotic masses, but no such adjustment is needed to keep all three phases present.

The resulting thermodynamic approximation is

$$\boxed{P_0(m) \simeq W_-\delta(m+m^*) + W_0\delta(m) + W_+\delta(m-m^*).} \tag{66}$$

The notation m here refers to the Hubbard–Stratonovich variable. The actual discrete magnetization $S(a)/N$ is a different finite-N random variable, though both concentrate at these same phase values.

B.3 Diffuse each branch: exact $P_t(x \mid m)$ and $P_t(m \mid x)$

Pass the clean spins through the same OU channel as in Appendix A:

$$P_t(x \mid a) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\Delta_t}} \exp\left[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\right], \quad \Delta_t = T(1-e^{-2t}). \tag{67}$$

We wish to calculate $P_t(x \mid m) = \sum_a P_t(x \mid a)P_0(a \mid m)$. Write

$$\zeta_i := \frac{e^{-t}x_i}{\Delta_t}, \qquad \kappa_t := \frac{e^{-2t}}{2\Delta_t}.$$

Expanding the Gaussian square reveals the key difference from Ising spins:

$$-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t} = -\frac{x_i^2}{2\Delta_t} + \zeta_i a_i - \kappa_t a_i^2. \tag{68}$$

For $a_i = \pm 1$ only, $a_i^2 = 1$ is constant. Here it is not, so the OU channel changes the relative likelihood of occupied and zero states. Define the one-coordinate Gaussian density $\varphi_\Delta(y) := (2\pi\Delta)^{-1/2}e^{-y^2/(2\Delta)}$. Summing over the three values of a_i gives

$$\begin{aligned} P_t(x_i \mid m) &= \varphi_{\Delta_t}(x_i)\frac{\sum_{a_i=-1,0,+1} e^{(u_m+\zeta_i)a_i - (\lambda+\kappa_t)a_i^2}}{D_\lambda(u_m)} \\ &= \varphi_{\Delta_t}(x_i)\frac{1 + 2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)}{1+2e^{-\lambda}\cosh u_m}. \end{aligned} \tag{69}$$

Conditional independence now yields the exact branch likelihood

$$\boxed{P_t(x \mid m) = \prod_{i=1}^{N}\varphi_{\Delta_t}(x_i)\prod_{i=1}^{N}\frac{1 + 2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)}{1+2e^{-\lambda}\cosh u_m}.} \tag{70}$$

Bayes' rule gives $P_t(m \mid x) \propto P_t(x \mid m)P_0(m)$. The factor $[D_\lambda(u_m)]^N$ from $P_0(m)$ cancels the denominators in (70); the common Gaussian factor cancels on normalization. Thus the exact latent posterior is

$$\boxed{P_t(m \mid x) = \frac{e^{-\beta N m^2/2}\displaystyle\prod_{i=1}^{N}[1 + 2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)]}{\displaystyle\int_{\mathbb{R}} \mathrm{d}v\, e^{-\beta N v^2/2}\prod_{i=1}^{N}[1 + 2e^{-(\lambda+\kappa_t)}\cosh(\beta v + h/N + \zeta_i)]}.} \tag{71}$$

This remains a one-dimensional integral over m, but its integrand generally depends on all of $x_1, \ldots, x_N$ separately. A scalar latent integration is not yet scalar *score closure*.

The same three-term sum gives the exact sitewise denoiser at fixed m:

$$\boxed{\mathbb{E}[a_i \mid x_i, m] = \frac{2e^{-(\lambda+\kappa_t)}\sinh(u_m+\zeta_i)}{1+2e^{-(\lambda+\kappa_t)}\cosh(u_m+\zeta_i)}.} \tag{72}$$

Averaging over (71) gives

$$\mathbb{E}[a_i \mid x] = \int_{\mathbb{R}} \mathrm{d}m\, \mathbb{E}[a_i \mid x_i, m] P_t(m \mid x), \tag{73}$$

and the universal exact OU identity yields

$$F_i(x,t) = -\frac{x_i}{\Delta_t} + \frac{e^{-t}}{\Delta_t}\mathbb{E}[a_i \mid x]. \tag{74}$$

This is a complete exact description, but it is not yet a closed function of the empirical mean.

B.4 High-noise expansion: why the branch evidence becomes scalar

We now use the three-delta approximation (66) and investigate times

$$t = \tfrac{1}{2}\log N + O(1), \qquad e^{-t} = O(N^{-1/2}), \quad \kappa_t = O(N^{-1}), \quad \zeta_i = O_{\mathbb{P}}(N^{-1/2}).$$

The conditional clean mean in branch α is $m_\alpha \in \{-m^*, 0, +m^*\}$. It is useful to introduce its occupancy and variance,

$$\begin{aligned} q_\alpha &:= \mathbb{E}[a_i^2 \mid m_\alpha] = \frac{2e^{-\lambda}\cosh(\beta m_\alpha)}{1+2e^{-\lambda}\cosh(\beta m_\alpha)}, \\ v_\alpha &:= \mathrm{Var}(a_i \mid m_\alpha) = q_\alpha - m_\alpha^2. \end{aligned} \tag{75}$$

Unlike in Ising Curie–Weiss, q_0 and $q_\pm$ need not be equal. We must therefore keep the quadratic likelihood terms instead of simply copying the two-branch likelihood-ratio expansion.

At fixed branch, divide its noisy likelihood by the common Gaussian baseline. The corresponding one-site factor is

$$L_\alpha(\zeta_i, \kappa_t) := \mathbb{E}\big[e^{\zeta_i a_i - \kappa_t a_i^2} \mid m_\alpha\big].$$

For bounded spins, Taylor expansion of its logarithm gives

$$\log L_\alpha(\zeta_i, \kappa_t) = m_\alpha \zeta_i - q_\alpha \kappa_t + \frac{v_\alpha}{2}\zeta_i^2 + O(|\zeta_i|^3 + \kappa_t|\zeta_i| + \kappa_t^2). \tag{76}$$

Summing over coordinates,

$$\log P_t(x \mid m_\alpha) = \sum_i \log \varphi_{\Delta_t}(x_i) + m_\alpha \sum_i \zeta_i - N q_\alpha \kappa_t + \frac{v_\alpha}{2}\sum_i \zeta_i^2 + o_{\mathbb{P}}(1). \tag{77}$$

At the chosen time scale, $N\kappa_t = O(1)$ and, under the noised law,

$$\frac{1}{N}\sum_i x_i^2 = \Delta_t + o_{\mathbb{P}}(1).$$

Consequently

$$\sum_i \zeta_i^2 = \frac{N e^{-2t}}{\Delta_t} + o_{\mathbb{P}}(1) = 2N\kappa_t + o_{\mathbb{P}}(1).$$

The occupation and variance terms therefore combine as

$$\begin{aligned} -N q_\alpha \kappa_t + \frac{v_\alpha}{2}\sum_i \zeta_i^2 &= -N\kappa_t(q_\alpha - v_\alpha) + o_{\mathbb{P}}(1) \\ &= -\frac{N e^{-2t}}{2\Delta_t} m_\alpha^2 + o_{\mathbb{P}}(1). \end{aligned} \tag{78}$$

The q_α-dependence cancels against the variance contribution. This cancellation is essential: the central and outer branches have different occupation statistics, yet no second collective observable is needed at leading order on this high-noise scale. We obtain

$$\boxed{\log P_t(x \mid m_\alpha) = C_t(x) + \frac{e^{-t}}{\Delta_t} m_\alpha \sum_i x_i - \frac{N e^{-2t}}{2\Delta_t} m_\alpha^2 + o_{\mathbb{P}}(1),} \tag{79}$$

where $C_t(x) = \sum_i \log \varphi_{\Delta_t}(x_i)$ is independent of the branch. The expansion is for typical noisy configurations; it is not an exact identity for every x.

Set

$$\mu := \frac{1}{\sqrt{N}} \sum_i x_i, \qquad c_t := m^* \sqrt{N} e^{-t}, \qquad \nu_t := \frac{c_t \mu}{\Delta_t}, \qquad d_t := \frac{c_t^2}{2\Delta_t}.$$

Combining (79) with the prior weights, the three branch posterior probabilities are

$$\boxed{\begin{aligned} p_0(\mu, t) &\simeq \frac{W_0}{D_t(\mu)}, \\ p_\pm(\mu, t) &\simeq \frac{W_\pm e^{\pm \nu_t - d_t}}{D_t(\mu)}, \\ D_t(\mu) &:= W_0 + e^{-d_t}[W_+ e^{\nu_t} + W_- e^{-\nu_t}]. \end{aligned}} \tag{80}$$

At $h = 0$, $W_+ = W_- = W$, so

$$\boxed{D_t(\mu) = W_0 + 2W e^{-d_t} \cosh(\nu_t).} \tag{81}$$

The factor e^{-d_t} is new relative to the symmetric *two*-branch calculation. It cannot be dropped: the central branch has squared mean zero while the outer branches have squared means $(m^*)^2$.

In each branch, (72) now gives $\mathbb{E}[a_i \mid x_i, m_\alpha] = m_\alpha + O(|\zeta_i| + \kappa_t)$. Averaging with (80),

$$\begin{aligned} \mathbb{E}[a_i \mid x] &\simeq -m^* p_-(\mu, t) + m^* p_+(\mu, t) \\ &= m^* \frac{2W e^{-d_t} \sinh(\nu_t)}{W_0 + 2W e^{-d_t} \cosh(\nu_t)} \qquad (h = 0). \end{aligned} \tag{82}$$

Thus the informative part of the full score becomes the same function of μ at every site, just as in binary Curie–Weiss, albeit with $\sinh / (W_0 + 2W \cosh)$ instead of a single tanh.

B.5 Projection and the candidate three-well potential

The increasing reverse clock is $s = t_f - t$. Project the full reverse OU SDE onto $\mu = N^{-1/2} \sum_i x_i$. The projected Brownian motion remains standard because

$$\frac{1}{N} \sum_{i,j} 2T \delta_{ij} \, \mathrm{d}s = 2T \, \mathrm{d}s.$$

Using (74) and the high-noise closed denoiser (82), the reverse drift becomes

$$\boxed{\begin{aligned} \mathrm{d}\mu_s &= b_3(\mu_s, t_f - s) \, \mathrm{d}s + \sqrt{2T} \, \mathrm{d}W_s^\mu, \\ b_3(\mu, t) &\simeq \left(1 - \frac{2T}{\Delta_t}\right) \mu + \frac{2T c_t}{\Delta_t} \frac{2W e^{-d_t} \sinh(\nu_t)}{W_0 + 2W e^{-d_t} \cosh(\nu_t)}. \end{aligned}} \tag{83}$$

This is now a scalar drift, and can be integrated. Since $\partial_\mu \nu_t = c_t / \Delta_t$, direct differentiation shows

$$\boxed{\begin{aligned} V_{3,\Delta}(\mu, t) = \frac{1}{2} \left(\frac{2T}{\Delta_t} - 1\right) \mu^2 \\ - 2T \log \left[W_0 + 2W e^{-c_t^2/(2\Delta_t)} \cosh\left(\frac{c_t \mu}{\Delta_t}\right)\right] + C(t), \end{aligned}} \tag{84}$$

with $b_3 = -\partial_\mu V_{3,\Delta}$ at the approximation order used. Only now, at large forward time, is it appropriate to set $\Delta_t \simeq T$. The particularly simple form is

$$\boxed{V_3(\mu,t) \simeq \frac{\mu^2}{2} - 2T\log\left[W_0 + 2We^{-c_t^2/(2T)}\cosh\left(\frac{c_t\mu}{T}\right)\right] + C(t).} \tag{85}$$

The additional constant-in-μ term *inside the logarithm* permits the central phase to compete with the two outer phases. Whether there are one or three visible wells depends on t, the phase separation, and the weights; three clean phases do not mean three reverse-time wells at every time.

B.6 Exact projected marginal: check and limitation of closure

There is an independent way to calculate a potential for μ, which is exact at every finite N. It also clarifies which part of the preceding argument is nontrivial. Unlike the auxiliary m, the *empirical* clean magnetization $S = \sum_i a_i$ takes integer values. At fixed $S = s$ let $q = Q = \sum_i a_i^2$ count occupied spins. Then

$$n_+ = \frac{q+s}{2}, \qquad n_- = \frac{q-s}{2}, \qquad n_0 = N - q.$$

The multinomial multiplicity and the occupancy cost give

$$C_N(s;\lambda) := \sum_{\substack{|s|\le q\le N \\ q\equiv s \pmod 2}} \frac{N!}{[(q+s)/2]![(q-s)/2]!(N-q)!} e^{-\lambda q}. \tag{86}$$

Hence the *exact* law of S is

$$\boxed{P_0(S=s) = \frac{C_N(s;\lambda)e^{\beta s^2/(2N)+hs/N}}{\displaystyle\sum_{r=-N}^{N} C_N(r;\lambda)e^{\beta r^2/(2N)+hr/N}}.} \tag{87}$$

Summing the forward OU coordinates gives, exactly,

$$\mu_t \mid S = s \sim \mathcal{N}\left(\frac{e^{-t}s}{\sqrt{N}}, \Delta_t\right). \tag{88}$$

Thus its one-dimensional density is the finite Gaussian sum

$$\boxed{P_t^\mu(\mu) = \sum_{s=-N}^{N} P_0(S=s)\,\varphi_{\Delta_t}\left(\mu - \frac{e^{-t}s}{\sqrt{N}}\right).} \tag{89}$$

Because μ_t is itself a forward OU process, its time reversal has an *exact* scalar potential

$$\boxed{V_{\text{marginal}}(\mu,t) = -\frac{\mu^2}{2} - 2T\log P_t^\mu(\mu) + C(t).} \tag{90}$$

This result does *not* require a three-phase approximation. At the speciation scale, concentration near m_α turns (89) into

$$P_t^\mu(\mu) \simeq W_-\varphi_{\Delta_t}(\mu + c_t) + W_0\varphi_{\Delta_t}(\mu) + W_+\varphi_{\Delta_t}(\mu - c_t),$$

which gives (84) by completing Gaussian squares. This is a useful independent check of the potential.

Marginal potential versus closure of the full score

For *any* clean prior, the linear coordinate $\mu = N^{-1/2}\sum_i x_i$ is itself an exact forward OU process, so its one-dimensional reversed marginal admits the potential (90). What is special about the present mean-field model is the stronger statement that the *projected full-state reverse drift*, evaluated at a microscopic configuration x, becomes a function of $\mu(x)$ alone at the high-noise scale. Equation (79) and the site-independent approximation in (82) are the ingredients that establish this latter, asymptotic closure.

Structural trend. The Ising case has two phases, while this three-state model has three coexisting phases when λ is tuned appropriately. Yet the high-noise branch evidence in both cases is controlled by the same scalar $\sum_i x_i$: the branches differ in their common mean along the uniform direction. The number of phases therefore need not equal the number of collective coordinates. The relevant requirement is that branch discrimination and the within-branch denoiser become functions of the proposed collective statistic, at the stated approximation order.

B.7 A first bridge toward n wells: number of phases versus latent dimension

The three-branch example suggests distinguishing the number of phases from the number of independent directions in which they differ. As a transparent reference case, consider K Gaussian branch laws with $X \mid B = \alpha \sim \mathcal{N}(v_\alpha, \Sigma)$, where the same positive-definite covariance Σ is used in each branch. Bayes' rule and expansion of the Gaussian exponents give, relative to branch 1,

$$\log \frac{P(B = \alpha \mid x)}{P(B = 1 \mid x)} = \log \frac{w_\alpha}{w_1} + x^\top \Sigma^{-1}(v_\alpha - v_1) - \frac{1}{2}\left(v_\alpha^\top \Sigma^{-1} v_\alpha - v_1^\top \Sigma^{-1} v_1\right).$$

Hence all posterior branch probabilities depend on x only through the linear projections onto the span of $\{\Sigma^{-1}(v_\alpha - v_1) : \alpha = 2, \dots, K\}$. The dimension of this span is at most $K - 1$, regardless of the ambient dimension. Several branches along one common direction can therefore be distinguished with just one scalar, whereas branches whose mean differences span several directions require several independent linear projections. This is a statement about *branch-posterior sufficiency*; full reverse-score closure requires the additional within-branch posterior means to depend only on those same projections, as shown for the two- and three-state mean-field models above. For non-Gaussian branch likelihoods this linear-sufficiency conclusion need not hold exactly.

For genuinely competing template fields $h^1, \dots, h^K$ on a common reference measure $\rho(\mathrm{d}a)$, the corresponding clean law is $P_0(\mathrm{d}a) = \sum_\alpha w_\alpha e^{\beta\langle a, h^\alpha\rangle}\rho(\mathrm{d}a)/Z_\alpha$. It must not be confused with multiplying all template factors:

$$e^{\beta \sum_\alpha \langle a, h^\alpha\rangle} = e^{\beta\langle a, \sum_\alpha h^\alpha\rangle},$$

which describes a single effective field, rather than a latent mixture. This elementary distinction underlies the outlook in the main text.

C Curie–Weiss reduction: complete derivation from the prior to the potential

Objective. This appendix presents the Curie–Weiss calculation as one continuous argument. Every approximation is introduced at the point where it first enters.

C.1 Clean prior, scaling, and Hubbard–Stratonovich decoupling

Start from

$$P_0(a) = \frac{1}{\mathcal{Z}_N} \exp\left[\frac{\beta}{2N}\left(\sum_i a_i\right)^2 + \frac{h}{N}\sum_i a_i\right].$$

If $m_a = N^{-1}\sum_i a_i$, then the interaction is $\beta N m_a^2/2$, so the $1/N$ factor makes it extensive. The field term is $\pm h m^*$ in the two ferromagnetic phases; it therefore creates a finite branch bias without suppressing one phase exponentially in N.

Let $S = \sum_i a_i$. Completing the square gives

$$-\frac{\beta N}{2}m^2 + \beta m S = -\frac{\beta N}{2}\left(m - \frac{S}{N}\right)^2 + \frac{\beta S^2}{2N}. \tag{91}$$

Using the Gaussian integral,

$$\exp\left(\frac{\beta S^2}{2N}\right) = \sqrt{\frac{\beta N}{2\pi}}\int_{\mathbb{R}} \mathrm{d}m\, \exp\left[-\frac{\beta N}{2}m^2 + \beta m S\right]. \tag{92}$$

Substitution into the clean law gives the exact augmented joint measure

$$P_0(m, a) = \frac{1}{\widetilde{\mathcal{Z}}_N}\exp\left(-\frac{\beta N}{2}m^2\right)\prod_{i=1}^N \exp[(\beta m + h/N)a_i]. \tag{93}$$

Integrating over m recovers the original $P_0(a)$. At fixed m, factors that depend only on m cancel from the conditional distribution, so

$$P_0(a \mid m) = \prod_{i=1}^N \frac{e^{(\beta m + h/N)a_i}}{2\cosh(\beta m + h/N)}. \tag{94}$$

Thus the interacting measure becomes a mixture of product measures.

C.2 Marginal magnetization law, phase concentration, and branch masses

Summing (93) over all spin configurations gives

$$\begin{aligned} P_0(m) &\propto \exp\left(-\frac{\beta N}{2}m^2\right)[2\cosh(\beta m + h/N)]^N \\ &\asymp \exp[-\beta N \Phi_N(m)], \end{aligned} \tag{95}$$

where, up to an additive constant independent of m,

$$\Phi_N(m) = \frac{m^2}{2} - \frac{1}{\beta}\log\cosh(\beta m + h/N).$$

Stationary points satisfy

$$m = \tanh(\beta m + h/N).$$

For $\beta > 1$ and large N, there are stable minima $m_\pm = \pm m^* + O(N^{-1})$, with $m^* = \tanh(\beta m^*)$.

The mass near either minimum is obtained by Laplace's method:

$$I_\pm := \int_{\text{near } m_\pm} e^{-\beta N \Phi_N(m)}\, \mathrm{d}m \simeq e^{-\beta N \Phi_N(m_\pm)} \sqrt{\frac{2\pi}{\beta N \Phi_N''(m_\pm)}}. \tag{96}$$

The curvature controls the peak width $O(N^{-1/2})$, while the exponential factor controls its mass. Expanding the weak field,

$$\Phi_N(m) = \Phi_0(m) - \frac{h}{\beta N}\tanh(\beta m) + O(N^{-2}),$$

so

$$e^{-\beta N \Phi_N(m_\pm)} \simeq e^{-\beta N \Phi_0(m^*)} e^{\pm h m^*}.$$

Since the two curvature factors agree at leading order,

$$I_+ : I_- = e^{h m^*} : e^{-h m^*},$$

and normalization gives (13). Therefore

$$P_0(m) \simeq W_+ \delta(m - m^*) + W_- \delta(m + m^*). \tag{97}$$

Conditional on branch $\pm$,

$$P_0(a \mid \pm) = \prod_{i=1}^N \frac{e^{\pm \beta m^* a_i}}{2\cosh(\beta m^*)}, \qquad \mathbb{E}[a_i \mid \pm] = \pm m^*. \tag{98}$$

C.3 Exact branch algebra after OU diffusion

For a single binary spin,

$$(x_i - e^{-t} a_i)^2 = x_i^2 + e^{-2t} - 2 x_i e^{-t} a_i, \tag{99}$$

so the spin-dependent part of the OU likelihood is $\exp(a_i \zeta_i)$ with

$$\zeta_i = \frac{x_i e^{-t}}{\Delta_t}.$$

Multiplying by the branch prior gives the local posterior field

$$A_i^\pm = \pm \beta m^* + \zeta_i.$$

The binary identity

$$\frac{\sum_{a=\pm 1} a e^{aA}}{\sum_{a=\pm 1} e^{aA}} = \tanh A$$

therefore yields

$$\mathbb{E}[a_i \mid x, \pm] = \tanh(\pm \beta m^* + \zeta_i). \tag{100}$$

Summing the clean spins inside each branch gives

$$\frac{\cosh(\pm \beta m^* + \zeta_j)}{\cosh(\beta m^*)} = \cosh \zeta_j \, [1 \pm m^* \tanh \zeta_j].$$

The $\cosh \zeta_j$ factors are common to both branches. Their branch-dependent likelihood factors are therefore exactly

$$Q_\pm(x) = \prod_j [1 \pm m^* \tanh \zeta_j],$$

and Bayes gives (15). The total posterior mean is

$$\mathbb{E}[a_i \mid x] = \frac{W_+ Q_+(x) \tanh(\beta m^* + \zeta_i) + W_- Q_-(x) \tanh(-\beta m^* + \zeta_i)}{W_+ Q_+(x) + W_- Q_-(x)}. \tag{101}$$

This is exact within the two-delta-peak thermodynamic model.

C.4 Controlled large-time closure

Assume $\max_i |\zeta_i| \ll 1$. Instead of separately approximating Q_+ and Q_-, consider their ratio:

$$\log \frac{Q_+}{Q_-} = 2 \sum_j \operatorname{artanh}(m^* \tanh \zeta_j)$$
$$= 2m^* \sum_j \zeta_j + O\left(\sum_j |\zeta_j|^3 \right). \tag{102}$$

Using $W_+/W_- = e^{2hm^*}$,

$$\log \frac{P(+ \mid x)}{P(- \mid x)} = 2m^* \left(h + NM(x) \frac{e^{-t}}{\Delta_t} \right) + O\left(\sum_j |\zeta_j|^3 \right). \tag{103}$$

Define

$$A(x,t) := m^* \left(h + NM(x) \frac{e^{-t}}{\Delta_t} \right).$$

Then

$$P(+ \mid x) - P(- \mid x) \simeq \tanh A(x,t).$$

At the same order,

$$\mathbb{E}[a_i \mid x, \pm] = \pm m^* + O(\zeta_i).$$

Substitution into (101) gives

$$\mathbb{E}[a_i \mid x] \simeq m^* \tanh A(x,t),$$

which is (17). It is independent of i and depends on x only through $M(x)$. The exact OU score identity now gives (18).

At the transition scale $t = \frac{1}{2} \log N + O(1)$, typical high-noise coordinates satisfy

$$\max_i |\zeta_i| = O_{\mathbb{P}}\left(\sqrt{\frac{\log N}{N}} \right),$$

so the coordinatewise expansion is self-consistent. The cubic remainder in (102) is then controlled under the same typical-event estimate.

C.5 Projection, exact-in-Δ_t potential, and large-time form

Define $\mu = N^{-1/2} \sum_i x_i$. The projected noise has covariance

$$\langle \eta_\mu(t) \eta_\mu(t') \rangle = \frac{1}{N} \sum_{i,j} 2T \delta_{ij} \delta(t - t') = 2T \delta(t - t'). \tag{104}$$

Summing the closed score gives

$$\frac{1}{\sqrt{N}} \sum_i F_i^{\mathrm{LT}}(x,t) = -\frac{\mu}{\Delta_t} + m^* \sqrt{N} \frac{e^{-t}}{\Delta_t} \tanh\left[m^* \left(h + \sqrt{N} \mu \frac{e^{-t}}{\Delta_t} \right) \right].$$

Substitution in the reverse equation gives (21). Direct differentiation verifies

$$-\partial_\mu V_\Delta = \left(1 - \frac{2T}{\Delta_t} \right) \mu + 2Tm^* \sqrt{N} \frac{e^{-t}}{\Delta_t} \tanh\left[m^* \left(h + \frac{\sqrt{N} e^{-t}}{\Delta_t} \mu \right) \right].$$

Thus (22) is the effective potential before the approximation $\Delta_t \simeq T$.

At large time, $\Delta_t = T(1 - e^{-2t}) = T[1 + o(1)]$, and the potential reduces to (26). In the increasing reverse clock, its minima are stable fixed points. For $h = 0$, the curvature at the origin is

$$V''(0,t) = 1 - \frac{2m^{*2} N}{T} e^{-2t},$$

which changes sign at (30). For smaller t, $\log \cosh Z \sim |Z|$ gives the two-well form (31).

D Spherical Curie–Weiss: complete derivation

Objective and route. We extend the preceding analysis by replacing the Ising configuration space with the sphere $\|a\|_2 = \sqrt{N}$, while leaving the quadratic Curie–Weiss interaction unchanged. The OU forward channel and reverse-score identity are taken from Appendix A, rather than rederived. The new work is to specify the reference measure, perform the Hubbard–Stratonovich augmentation, evaluate its spherical integral with the complete surface Jacobian, and follow the resulting posterior through diffusion to the projected potential. The derivation keeps three objects separate throughout:

$$q(a) = \frac{1}{N}\sum_i a_i \in [-1,1], \qquad m \in \mathbb{R}, \qquad \mu(x) = \frac{1}{\sqrt{N}}\sum_i x_i.$$

Here q is the physical clean magnetization, m is the auxiliary Hubbard–Stratonovich variable, and μ is the collective coordinate of the noisy observation. A density denoted p_0^q refers to q; a density denoted p_0^{HS} refers to m. The superscripts identify the random variable, not a different clean model. We work with $N \geq 3$. Thermodynamic two-phase and high-noise statements later require fixed $\beta > 1$ and fixed h, T with $T > 0$.

D.1 The clean law: unchanged interaction, different reference measure

The Ising restriction $a_i \in \{-1, +1\}$ fixes every coordinate separately. It implies $\|a\|_2^2 = N$. Our new restriction fixes only that total norm:

$$\mathbb{S}^{N-1}_{\sqrt{N}} := \{a \in \mathbb{R}^N : \|a\|_2^2 = N\}.$$

The superscript records the dimension $N-1$ of the surface; the ambient space still has dimension N. Individual coordinates are continuous and need not have magnitude one. The symmetry makes their mean squared value one under the normalized uniform measure, but it does not bound each coordinate independently by one.

Let σ_N be normalized uniform surface measure on this sphere and set

$$\begin{aligned} P_0^{\mathrm{sph}}(\,\mathrm{d}a) &= \frac{1}{Z_N^{\mathrm{sph}}} e^{\frac{\beta}{2N}S(a)^2 + \frac{h}{N}S(a)}\,\mathrm{d}\sigma_N(a), \\ S(a) &:= \sum_{i=1}^N a_i, \qquad Z_N^{\mathrm{sph}} := \int_{\mathbb{S}^{N-1}_{\sqrt{N}}} e^{\frac{\beta}{2N}S(a)^2 + \frac{h}{N}S(a)}\,\mathrm{d}\sigma_N(a). \end{aligned} \tag{105}$$

The partition function Z_N^{sph} is finite because the sphere is compact. The parameters β and h play exactly the same roles as before. In particular, the square is *outside* the sum:

$$S(a)^2 = \Big(\sum_i a_i\Big)^2 = \sum_i a_i^2 + 2\sum_{i<j} a_i a_j.$$

The first term equals N, but the cross terms are not constant. Replacing $S(a)^2$ by $\sum_i a_i^2$ would remove the interaction and would define a different model.

For a measurable region E on the sphere, the meaning of (105) is

$$P_0^{\mathrm{sph}}(E) = \frac{1}{Z_N^{\mathrm{sph}}}\int_E e^{\frac{\beta}{2N}S(a)^2 + \frac{h}{N}S(a)}\,\mathrm{d}\sigma_N(a).$$

The exponential supplies a relative density with respect to σ_N. It does not assign a positive mass to an individual point. In the discrete Ising calculation, counting measure was implicit in the sum over configurations; here surface measure must be specified explicitly. Ordinary N-dimensional Lebesgue volume cannot be restricted by merely integrating over the sphere, since the sphere has zero ambient volume.

Norm and thermodynamic scaling. Since $S = Nq$, the exponent is

$$\frac{\beta}{2N}S^2 + \frac{h}{N}S = \frac{\beta N}{2}q^2 + hq.$$

For a phase with q of order one, the interaction is of order N, or order one *per coordinate*. The weak field is of order one in total. This matches the Ising normalization. Under the uniform reference measure, q instead fluctuates on a smaller scale, so the interaction need not be order N for every individual configuration. Choosing radius one with the same coefficient $\beta/(2N)$ would not preserve the magnetized-state scaling: Cauchy–Schwarz would bound S^2 by N, making the interaction at most order one.

D.2 Hubbard–Stratonovich augmentation, with all constants retained

The Gaussian identity is independent of the configuration space. For a fixed real number S and $\beta > 0$, complete the square:

$$\begin{aligned} -\frac{\beta N}{2}m^2 + \beta m S &= -\frac{\beta N}{2}\left(m^2 - 2\frac{S}{N}m\right) \\ &= -\frac{\beta N}{2}\left(m - \frac{S}{N}\right)^2 + \frac{\beta S^2}{2N}. \end{aligned} \tag{106}$$

The term independent of m factors out of the integral. With $z = \sqrt{\beta N}(m - S/N)$, the differential is $\mathrm{d}m = \mathrm{d}z/\sqrt{\beta N}$, and therefore

$$\begin{aligned} \int_{\mathbb{R}} e^{-\beta N m^2/2 + \beta m S}\,\mathrm{d}m &= e^{\beta S^2/(2N)}\int_{\mathbb{R}} e^{-\frac{\beta N}{2}(m-S/N)^2}\,\mathrm{d}m \\ &= e^{\beta S^2/(2N)}\frac{1}{\sqrt{\beta N}}\int_{\mathbb{R}} e^{-z^2/2}\,\mathrm{d}z = e^{\beta S^2/(2N)}\sqrt{\frac{2\pi}{\beta N}}. \end{aligned} \tag{107}$$

Thus

$$e^{\beta S^2/(2N)} = \sqrt{\frac{\beta N}{2\pi}}\int_{\mathbb{R}} e^{-\beta N m^2/2 + \beta m S}\,\mathrm{d}m. \tag{108}$$

Substituting $S = S(a)$ into the clean law identifies the exact joint measure

$$\boxed{P_0^{\mathrm{sph}}(\,\mathrm{d}m,\,\mathrm{d}a) = \frac{1}{Z_N^{\mathrm{sph}}}\sqrt{\frac{\beta N}{2\pi}}e^{-\beta N m^2/2 + (\beta m + h/N)S(a)}\,\mathrm{d}m\;\mathrm{d}\sigma_N(a).} \tag{109}$$

Integrating over m recovers (105); hence the augmentation has not changed the physical distribution of a. The factorization $e^{(\beta m + h/N)S(a)} = \prod_i e^{(\beta m + h/N)a_i}$ still holds algebraically, but the *reference measure* is not a product measure. The constraint $\sum_i a_i^2 = N$ continues to couple the coordinates. Conditional independence does not follow from this product of exponentials.

Define the field $\theta_m := \beta m + h/N$ and the spherical partition integral

$$\mathcal{J}_N(v) := \int_{\mathbb{S}^{N-1}_{\sqrt{N}}} e^{v\cdot a}\,\mathrm{d}\sigma_N(a), \qquad v \in \mathbb{R}^N. \tag{110}$$

Writing $\mathbf{1} = (1, \dots, 1)$, the conditional law and the auxiliary density are consequently

$$P_0^{\mathrm{sph}}(\,\mathrm{d}a \mid m) = \frac{e^{\theta_m S(a)}}{\mathcal{J}_N(\theta_m \mathbf{1})}\,\mathrm{d}\sigma_N(a), \tag{111}$$

$$p_0^{\mathrm{HS}}(m) = \frac{1}{Z_N^{\mathrm{sph}}}\sqrt{\frac{\beta N}{2\pi}}e^{-\beta N m^2/2}\mathcal{J}_N(\theta_m \mathbf{1}). \tag{112}$$

This is exactly the same marginalization step as the Ising calculation. The remaining task is to evaluate the integral in $\mathcal{J}_N$, not to replace it by an unjustified product of one-site factors.

D.3 The spherical change of variables: vector differential, metric, and surface Jacobian

The integral in (112) depends on a only through $S(a)$. We therefore choose coordinates adapted to that sum. Define

$$u := \frac{1}{\sqrt{N}}(1,\dots,1), \qquad \|u\| = 1, \qquad q(a) := \frac{S(a)}{N} = \frac{u\cdot a}{\sqrt{N}}.$$

The vector u spans one direction, and its orthogonal complement $u^\perp$ has dimension $N-1$. Orthogonal projection gives

$$a = (u\cdot a)u + a_\perp = \sqrt{N}q\,u + a_\perp, \qquad u\cdot a_\perp = 0. \tag{113}$$

Taking squared norms and using orthogonality,

$$N = \|a\|^2 = Nq^2 + \|a_\perp\|^2, \qquad \|a_\perp\| = \sqrt{N(1-q^2)}.$$

For $-1 < q < 1$, let $\omega := a_\perp/\|a_\perp\|$. Then $\omega \in u^\perp$ and $\|\omega\| = 1$; it lies on an $(N-2)$-dimensional unit sphere, not on all of $u^\perp$. The norm constraint removes one of the $N-1$ coordinates in that subspace. Set $q = \cos\theta$ with $0 < \theta < \pi$. The parameterization is

$$a = \sqrt{N}\big(\cos\theta\, u + \sin\theta\,\omega\big). \tag{114}$$

There is one polar coordinate θ and $N-2$ coordinates describing ω, for a total of $N-1$ surface coordinates. At $\theta = 0, \pi$ the transverse direction is not uniquely defined; these poles have zero surface measure and do not affect the integrals.

The vector differential is not an area element. As both θ and ω change, the total differential is

$$\mathrm{d}a = \sqrt{N}(-\sin\theta\, u + \cos\theta\,\omega)\,\mathrm{d}\theta + \sqrt{N}\sin\theta\ \mathrm{d}\omega. \tag{115}$$

This is an infinitesimal vector displacement in $\mathbb{R}^N$. Differentiating $\omega\cdot\omega = 1$ and $u\cdot\omega = 0$, with fixed u, gives $\omega\cdot\mathrm{d}\omega = 0$ and $u\cdot\mathrm{d}\omega = 0$. The two terms in (115) are therefore orthogonal. Their squared lengths give

$$\boxed{\|\mathrm{d}a\|^2 = N\ \mathrm{d}\theta^2 + N\sin^2\theta\,\|\mathrm{d}\omega\|^2.} \tag{116}$$

This is a distance formula. To obtain surface area, we must calculate the volume spanned by all independent tangent directions, rather than treat $\mathrm{d}a$ itself as a scalar integration element.

Define the mapping and its derivative. Use local coordinates $\varphi = (\varphi_1,\dots,\varphi_{N-2})$ on the transverse unit sphere, so that $\omega = \omega(\varphi)$. Define

$$\mathcal{P}(\theta,\varphi) := \sqrt{N}[\cos\theta\, u + \sin\theta\,\omega(\varphi)] \quad \text{and} \quad \xi = (\theta,\varphi_1,\dots,\varphi_{N-2}). \tag{117}$$

Thus $\mathcal{P}$ maps $N-1$ coordinate parameters into N ambient coordinates. Its derivative $D\mathcal{P}$ is the $N\times(N-1)$ matrix whose columns are the tangent vectors

$$D\mathcal{P} = \big[\partial_\theta\mathcal{P}\ \ \partial_{\varphi_1}\mathcal{P}\ \ \cdots\ \ \partial_{\varphi_{N-2}}\mathcal{P}\big].$$

It has no ordinary determinant because it is rectangular. The associated square Gram matrix and the surface Jacobian are

$$g := (D\mathcal{P})^\top D\mathcal{P}, \qquad g_{ij} := \partial_{\xi_i}\mathcal{P}\cdot\partial_{\xi_j}\mathcal{P}, \qquad J_{\mathcal{P}} := \sqrt{\det g}. \tag{118}$$

Why this formula? A coordinate rectangle with side widths $\mathrm{d}\xi_i$ maps, to first order, to a tangent parallelepiped with edge vectors $(\partial_{\xi_i}\mathcal{P})\,\mathrm{d}\xi_i$. The squared volume of a parallelepiped is the determinant of the Gram matrix of its edges. Taking the square root therefore converts coordinate volume into surface area:

$$\mathrm{d}A = J_{\mathcal{P}}(\xi)\ \mathrm{d}\theta\ \mathrm{d}\varphi_1\cdots\mathrm{d}\varphi_{N-2}. \tag{119}$$

The differentials are the side widths of the coordinate patch; the Jacobian is the conversion factor, not another differential.

Calculate the metric and determinant explicitly. From (117),

$$\begin{aligned}\partial_\theta \mathcal{P} &= \sqrt{N}(-\sin\theta\, u + \cos\theta\, \omega),\\ \partial_{\varphi_j} \mathcal{P} &= \sqrt{N}\sin\theta\, \partial_{\varphi_j}\omega.\end{aligned} \tag{120}$$

Define the transverse metric $\gamma_{ij} := \partial_{\varphi_i}\omega \cdot \partial_{\varphi_j}\omega$. The constraints imply $u \cdot \partial_{\varphi_j}\omega = 0$ and $\omega \cdot \partial_{\varphi_j}\omega = 0$, so

$$g_{\theta\theta} = N, \qquad g_{\theta j} = 0, \qquad g_{ij} = N\sin^2\theta\, \gamma_{ij}$$

for angular indices i, j. Equivalently,

$$g = \begin{pmatrix} N & 0 \\ 0 & N\sin^2\theta\, \gamma \end{pmatrix}. \tag{121}$$

Since γ has size $(N-2)\times(N-2)$,

$$\begin{aligned}\det g &= N\det(N\sin^2\theta\,\gamma) = N(N\sin^2\theta)^{N-2}\det\gamma,\\ \sqrt{\det g} &= N^{(N-1)/2}(\sin\theta)^{N-2}\sqrt{\det\gamma}.\end{aligned} \tag{122}$$

Here $\sin\theta \geq 0$ on the chosen polar range. The transverse unit-sphere area element is, by the same rule,

$$\mathrm{d}\Omega_{N-2}(\omega) := \sqrt{\det\gamma}\ \mathrm{d}\varphi_1 \cdots \mathrm{d}\varphi_{N-2}.$$

Thus

$$\boxed{\mathrm{d}A = N^{(N-1)/2}(\sin\theta)^{N-2}\,\mathrm{d}\theta\ \mathrm{d}\Omega_{N-2}(\omega).} \tag{123}$$

There is one length factor $\sqrt{N}$ from the polar direction and $N-2$ factors $\sqrt{N}\sin\theta$ from the transverse directions. This provides an equivalent geometric interpretation of the determinant calculation.

A three-dimensional check. For $N = 3$, rotate coordinates so that $u = (0,0,1)$ and let $\omega(\varphi) = (\cos\varphi, \sin\varphi, 0)$. Then

$$\mathcal{P}(\theta,\varphi) = \sqrt{3}(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \qquad \gamma = (1), \qquad g = \begin{pmatrix} 3 & 0 \\ 0 & 3\sin^2\theta \end{pmatrix}.$$

Hence $J_{\mathcal{P}} = 3\sin\theta$ and $\mathrm{d}A = 3\sin\theta\ \mathrm{d}\theta\ \mathrm{d}\varphi$. This is the usual $R^2\sin\theta$ surface Jacobian with $R = \sqrt{3}$. The use of a vertical u is just a rotation of the original uniform magnetization direction, not a change of the model.

Now change the polar coordinate to the magnetization. The equality $q = \cos\theta$ gives

$$\mathrm{d}q = -\sin\theta\ \mathrm{d}\theta, \qquad \mathrm{d}\theta = -\frac{\mathrm{d}q}{\sqrt{1-q^2}}.$$

Consequently,

$$\begin{aligned}(\sin\theta)^{N-2}\,\mathrm{d}\theta &= -(1-q^2)^{(N-2)/2}\frac{\mathrm{d}q}{\sqrt{1-q^2}}\\ &= -(1-q^2)^{(N-3)/2}\,\mathrm{d}q.\end{aligned} \tag{124}$$

The limits $\theta : 0 \to \pi$ correspond to $q : 1 \to -1$. Reversing them removes the minus sign. In positive measure notation,

$$\boxed{\mathrm{d}A = N^{(N-1)/2}(1-q^2)^{(N-3)/2}\,\mathrm{d}q\ \mathrm{d}\Omega_{N-2}(\omega), \qquad -1 < q < 1.} \tag{125}$$

The exponent $(N-3)/2$ contains two contributions: $(N-2)/2$ from transverse area, and $-1/2$ from the differential substitution. Using transverse cross-sectional area alone would miss this last factor.

Normalize the area to uniform probability measure. Let $\mathcal{S}_k := 2\pi^{(k+1)/2}/\Gamma((k+1)/2)$ be the area of the unit k-sphere. Here the gamma function is $\Gamma(z) := \int_0^\infty y^{z-1}e^{-y}\,\mathrm{d}y$ for $z > 0$. For completeness, the area formula follows by evaluating the Gaussian integral in d dimensions in Cartesian and radial coordinates:

$$\pi^{d/2} = \int_{\mathbb{R}^d} e^{-\|y\|^2}\,\mathrm{d}^d y = \mathcal{S}_{d-1}\int_0^\infty e^{-r^2} r^{d-1}\,\mathrm{d}r = \frac{\mathcal{S}_{d-1}}{2}\Gamma(d/2).$$

In the last step, $v = r^2$ gives $r^{d-1}\,\mathrm{d}r = \frac{1}{2}v^{d/2-1}\,\mathrm{d}v$. The total area of our sphere is $N^{(N-1)/2}\mathcal{S}_{N-1}$. Define $\mathrm{d}\bar{\sigma}_{N-2} := \mathrm{d}\Omega_{N-2}/\mathcal{S}_{N-2}$. Dividing (125) by the total area gives

$$\boxed{\begin{aligned}\mathrm{d}\sigma_N(a) &= C_N(1-q^2)^{(N-3)/2}\,\mathrm{d}q\;\mathrm{d}\bar{\sigma}_{N-2}(\omega),\\ C_N &:= \frac{\mathcal{S}_{N-2}}{\mathcal{S}_{N-1}} = \frac{\Gamma(N/2)}{\sqrt{\pi}\,\Gamma((N-1)/2)}.\end{aligned}} \tag{126}$$

The radius factors cancel because the measure is normalized. As a check,

$$\int_{-1}^{1}(1-q^2)^{(N-3)/2}\,\mathrm{d}q = B\left(\tfrac{1}{2},\tfrac{N-1}{2}\right) = C_N^{-1}.$$

The beta function $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ follows here by substituting $v = q^2$ on each half of the interval. For $N = 3$, $C_3 = 1/2$ and the uniform-sphere magnetization is uniform on $(-1,1)$, as the three-dimensional check predicts.

Four different meanings of a differential

In the calculation above, $\mathrm{d}a$ is a vector displacement, $\|\mathrm{d}a\|^2$ is its squared length, $\mathrm{d}A$ is surface area, and $\mathrm{d}\sigma_N(a) = \mathrm{d}A/[N^{(N-1)/2}\mathcal{S}_{N-1}]$ is normalized surface probability. None is the ambient volume element $\mathrm{d}^N a = \mathrm{d}a_1\cdots\mathrm{d}a_N$. In measure notation such as $P_0(\mathrm{d}a)$, the symbol $\mathrm{d}a$ instead identifies the integration variable of a probability measure; it is not the tangent vector from (115). The notation is conventional, but the roles are different.

D.4 Evaluate the spherical integral and distinguish the two magnetization laws

We can now return to the exact starting point (112). In the coordinates just derived, $S(a) = Nq$. Substitute both the integrand and its measure:

$$\begin{aligned}\mathcal{J}_N(\theta_m\mathbf{1}) &= C_N\int_{-1}^{1}\int_{S^{N-2}} e^{N\theta_m q}(1-q^2)^{(N-3)/2}\,\mathrm{d}\bar{\sigma}_{N-2}(\omega)\;\mathrm{d}q\\ &= C_N\int_{-1}^{1} e^{N\beta mq+hq}(1-q^2)^{(N-3)/2}\,\mathrm{d}q.\end{aligned} \tag{127}$$

The angular integral is one because its measure is normalized and the integrand does not depend on ω. Thus

$$\boxed{p_0^{\mathrm{HS}}(m) = \frac{C_N}{Z_N^{\mathrm{sph}}}\sqrt{\frac{\beta N}{2\pi}}\int_{-1}^{1}(1-q^2)^{(N-3)/2} \times e^{-\beta Nm^2/2+\beta Nmq+hq}\,\mathrm{d}q, \qquad m\in\mathbb{R}.} \tag{128}$$

This is an exact one-dimensional integral, not a thermodynamic approximation. In Ising, the corresponding sum gave $[2\cosh(\beta m + h/N)]^N$; the spherical constraint instead gives (127).

Completing the square a second time,

$$-\frac{\beta N}{2}m^2 + \beta Nmq = -\frac{\beta N}{2}(m-q)^2 + \frac{\beta N}{2}q^2.$$

The partition function and the physical magnetization density are

$$Z_N^{\mathrm{sph}} = C_N \int_{-1}^{1} (1-q^2)^{(N-3)/2} e^{\beta N q^2/2 + hq}\, \mathrm{d}q, \tag{129}$$

$$p_0^q(q) = \frac{C_N}{Z_N^{\mathrm{sph}}} (1-q^2)^{(N-3)/2} e^{\beta N q^2/2 + hq}, \qquad -1 < q < 1. \tag{130}$$

Substitution into (128) gives the exact relation

$$\boxed{p_0^{\mathrm{HS}}(m) = \int_{-1}^{1} p_0^q(q) \sqrt{\frac{\beta N}{2\pi}} e^{-\beta N (m-q)^2/2}\, \mathrm{d}q.} \tag{131}$$

Equivalently,

$$m \mid a \sim \mathcal{N}\left(q(a), \frac{1}{\beta N}\right). \tag{132}$$

The auxiliary m is the physical magnetization plus an independent Gaussian augmentation of variance $1/(\beta N)$. Integrating the Gaussian in (131) over all m gives one, so the formula also checks the normalization of the HS density. At finite N, the support of q is $[-1, 1]$, whereas the support of m is all of $\mathbb{R}$.

The binary counterpart and why the document uses a different formula. For Ising spins, $q = S/N$ takes the discrete values $-1, -1 + 2/N, \ldots, 1$. There are $\binom{N}{N(1+q)/2}$ configurations with magnetization q. Consequently its exact mass function is

$$P_0^{\mathrm{I}}(q) = \frac{1}{Z_N^{\mathrm{I}}} \binom{N}{N(1+q)/2} e^{\beta N q^2/2 + hq}.$$

The auxiliary Ising density in Appendix C.2 is instead

$$p_0^{\mathrm{I,HS}}(m) \propto e^{-\beta N m^2/2} [2\cosh(\beta m + h/N)]^N.$$

These are not competing derivations of the same finite-N random variable. The same conditional Gaussian (132) relates them, with the integral over q in (131) replaced by a discrete sum. In both geometries, one must distinguish empirical magnetization from auxiliary magnetization before comparing their densities or widths.

Optional special-function expression. Define the normalized angular transform

$$\mathcal{A}_d(z) := \int_{S^{d-1}} e^{z\omega_1}\, \mathrm{d}\bar{\sigma}_{d-1}(\omega), \qquad \mathcal{A}_d(0) = 1. \tag{133}$$

Here ω_1 is the component of a uniform unit vector in any fixed coordinate direction of $\mathbb{R}^d$; rotation invariance makes the choice irrelevant. The same one-coordinate integration as above gives

$$\mathcal{A}_d(z) = \Gamma(d/2) \left(\frac{2}{|z|}\right)^{d/2-1} I_{d/2-1}(|z|), \qquad z \neq 0, \tag{134}$$

where I_ν is the modified Bessel function of the first kind, of order ν. The value at zero is supplied by continuity. This follows from the integral representation of I_ν with $\nu = d/2 - 1$; see DLMF 10.32.2. In particular,

$$\mathcal{J}_N(v) = \mathcal{A}_N(\sqrt{N}\|v\|), \qquad \mathcal{J}_N(\theta_m \mathbf{1}) = \mathcal{A}_N(N\theta_m),$$

where the last equality uses the evenness of $\mathcal{A}_N$. Equation (112) is therefore also an exact special-function expression. The integral representation remains more transparent for the thermodynamic and posterior calculations below.

D.5 Thermodynamic phases and their probabilities: physical and HS routes

Write the physical magnetization density as

$$p_0^q(q) = \frac{C_N}{Z_N^{\mathrm{sph}}}(1-q^2)^{-3/2}\exp[Nf_\beta(q)+hq], \qquad f_\beta(q) := \frac{\beta}{2}q^2 + \frac{1}{2}\log(1-q^2). \tag{135}$$

This is an exact rearrangement on $(-1,1)$. The phase locations maximize f_β, since it is multiplied by the large parameter N in the exponent. The weak field and the prefactor affect subleading shifts and branch weights. Equivalently one can minimize $\Phi_{\mathrm{sph}}(q) := -f_\beta(q)/\beta$.

Differentiate explicitly:

$$\begin{aligned} f'_\beta(q) &= \beta q - \frac{q}{1-q^2} = q\left(\beta - \frac{1}{1-q^2}\right), \\ f''_\beta(q) &= \beta - \frac{1+q^2}{(1-q^2)^2}. \end{aligned} \tag{136}$$

Thus $q=0$ is always stationary. For $\beta > 1$, two additional solutions are

$$m^*_{\mathrm{sph}} := \sqrt{1-\beta^{-1}}, \qquad q_\pm = \pm m^*_{\mathrm{sph}}. \tag{137}$$

At the origin, $f''_\beta(0) = \beta - 1 > 0$, so the origin is not a dominant maximum. At the two nonzero solutions,

$$f''_\beta(\pm m^*_{\mathrm{sph}}) = -2\beta(\beta-1) < 0. \tag{138}$$

Since f_β tends to $-\infty$ at the endpoints, the two nonzero maxima are the dominant phases. These are leading thermodynamic locations; the finite-N modes of the exact density need not be exactly equal to them. For $0 < \beta < 1$ there is only the central maximum. At $\beta = 1$ the quadratic term vanishes and the leading term near zero is $-q^4/4$, so the usual Gaussian $N^{-1/2}$ peak-width argument no longer applies. Our subsequent two-phase calculation keeps $\beta > 1$ fixed, rather than letting it approach this critical value with N.

Width is not probability mass. Let $\varepsilon \in \{-1,+1\}$ label a branch and choose a small neighborhood of $\varepsilon m^*_{\mathrm{sph}}$ that excludes the other branch. Its unnormalized integrated mass is

$$I_{\varepsilon,N} := \int_{\text{near } \varepsilon m^*_{\mathrm{sph}}} (1-q^2)^{-3/2} e^{Nf_\beta(q)+hq}\,\mathrm{d}q.$$

Taylor expansion at the maximum gives

$$f_\beta(q) = f_\beta(m^*_{\mathrm{sph}}) - \beta(\beta-1)(q-\varepsilon m^*_{\mathrm{sph}})^2 + O((q-\varepsilon m^*_{\mathrm{sph}})^3).$$

Factoring out the height and evaluating the local Gaussian integral yields

$$I_{\varepsilon,N} \sim (1-(m^*_{\mathrm{sph}})^2)^{-3/2} e^{Nf_\beta(m^*_{\mathrm{sph}})+\varepsilon h m^*_{\mathrm{sph}}} \sqrt{\frac{2\pi}{2N\beta(\beta-1)}}. \tag{139}$$

The common height, geometric prefactor, and width cancel between the two branches. The only surviving asymmetry is $e^{\varepsilon h m^*_{\mathrm{sph}}}$. Therefore

$$\boxed{W_\varepsilon = \frac{e^{\varepsilon h m^*_{\mathrm{sph}}}}{2\cosh(h m^*_{\mathrm{sph}})}, \qquad p_0^q \implies W_-\delta_{-m^*_{\mathrm{sph}}} + W_+\delta_{m^*_{\mathrm{sph}}}.} \tag{140}$$

The arrow denotes convergence of probability measures as $N \to \infty$. At $h = 0$, symmetry makes the two branch masses equal. With fixed nonzero h, both remain nonvanishing in this limit. The Gaussian variance of the physical magnetization within either branch is, to leading order,

$$\mathrm{Var}(q \mid \varepsilon) \sim \frac{1}{2\beta(\beta-1)N}. \tag{141}$$

The peak-height factor may be absorbed when describing the shape of a *single normalized peak*; it must be retained when comparing the masses of different peaks.

Recover the same phases directly from the HS integral. The joint (m, q) density from (128) has leading exponent $N\mathcal{L}_\beta(m, q)$, where

$$\mathcal{L}_\beta(m, q) := -\frac{\beta}{2}m^2 + \beta mq + \frac{1}{2}\log(1 - q^2). \tag{142}$$

The factor $e^{hq}(1 - q^2)^{-3/2}$ is outside this leading exponent. The two stationarity equations are

$$\partial_m\mathcal{L}_\beta = -\beta m + \beta q = 0, \qquad \partial_q\mathcal{L}_\beta = \beta m - \frac{q}{1 - q^2} = 0.$$

The first gives $m = q$ *at the saddle*. The second then becomes the physical stationarity equation in (136). At each nonzero saddle the Hessian is

$$D^2\mathcal{L}_\beta = \begin{pmatrix} -\beta & \beta \\ \beta & -\beta(2\beta - 1) \end{pmatrix}, \qquad \det D^2\mathcal{L}_\beta = 2\beta^2(\beta - 1) > 0.$$

Its first diagonal entry is negative, so it is negative definite. Both nonzero saddles are maxima. This reproduces the two phases by the same latent-variable route used in Appendix C.2.

The exact Gaussian relation (131) also shows why the auxiliary and physical variables have the same phase weights but different within-phase widths:

$$\mathrm{Var}(m \mid \varepsilon) = \mathrm{Var}(q \mid \varepsilon) + \frac{1}{\beta N} \sim \frac{2\beta - 1}{2\beta(\beta - 1)N}. \tag{143}$$

Here the branch event is defined from the physical magnetization. It agrees asymptotically with the sign of the auxiliary variable away from the critical regime. The equality $m = q$ should never be imposed inside the exact finite-N probabilities; it is a statement about their dominant saddles.

D.6 The OU likelihood and the exact latent posteriors

Take the forward formulas from Appendix A.1 as given:

$$P_t(x \mid a) = (2\pi\Delta_t)^{-N/2}e^{-\|x - e^{-t}a\|^2/(2\Delta_t)}, \qquad \Delta_t = T(1 - e^{-2t}), \quad t > 0.$$

Both Ising and spherical configurations have fixed squared norm N. Thus

$$\|x - e^{-t}a\|^2 = \|x\|^2 + Ne^{-2t} - 2e^{-t}x \cdot a.$$

Define the scalar $\ell_t := e^{-t}/\Delta_t$ and the common likelihood factor

$$K_t(x) := (2\pi\Delta_t)^{-N/2}\exp\left[-\frac{\|x\|^2 + Ne^{-2t}}{2\Delta_t}\right]. \tag{144}$$

Then $P_t(x \mid a) = K_t(x)e^{\ell_t x \cdot a}$. Unlike the three-state model, there is no configuration-dependent $\|a\|^2$ penalty left after this expansion.

Condition first on the HS variable. Multiply this likelihood by (111) and integrate over a:

$$\boxed{P_t(x \mid m) = K_t(x)\frac{\mathcal{J}_N(\theta_m\mathbf{1} + \ell_t x)}{\mathcal{J}_N(\theta_m\mathbf{1})}.} \tag{145}$$

Bayes' rule multiplies this expression by $p_0^{\mathrm{HS}}(m) \propto e^{-\beta Nm^2/2}\mathcal{J}_N(\theta_m\mathbf{1})$. The denominator cancels, and $K_t(x)$ is independent of m. After normalization,

$$\boxed{p_t^{\mathrm{HS}}(m \mid x) = \frac{e^{-\beta Nm^2/2}\mathcal{J}_N((\beta m + h/N)\mathbf{1} + \ell_t x)}{\displaystyle\int_{\mathbb{R}} e^{-\beta Nv^2/2}\mathcal{J}_N((\beta v + h/N)\mathbf{1} + \ell_t x)\,\mathrm{d}v}.} \tag{146}$$

This is the exact spherical counterpart of the Ising expression $e^{-\beta Nm^2/2}\prod_i 2\cosh(\beta m + h/N + \zeta_i)$. There is no product factorization, but there is still just one HS integration.

Condition instead on the physical magnetization. The same model admits a second exact formula that is more transparent for closure. Decompose the observation as

$$x = \mu u + x_\perp, \quad \mu = u \cdot x, \quad x_\perp = x - \mu u, \quad r = \|x_\perp\|,$$

and define $R_q := \sqrt{N(1-q^2)}$. Under the clean law conditional on q, the vector $a_\perp = R_q\omega$ has uniform orientation in $u^\perp$: the Gibbs factor depends only on q and therefore does not tilt that orientation. Since

$$x \cdot a = \sqrt{N}q\mu + R_q x_\perp \cdot \omega,$$

angular integration with (133) gives

$$\boxed{P_t(x \mid q) = K_t(x) e^{\ell_t \sqrt{N} q\mu} \mathcal{A}_{N-1}(\ell_t r R_q).} \tag{147}$$

This is exact, including its normalization as a density in x. Consequently,

$$\boxed{p_t^q(q \mid x) = \frac{p_0^q(q) e^{\ell_t \sqrt{N} q\mu} \mathcal{A}_{N-1}(\ell_t r R_q)}{\displaystyle\int_{-1}^{1} p_0^q(v) e^{\ell_t \sqrt{N} v\mu} \mathcal{A}_{N-1}(\ell_t r R_v)\, \mathrm{d}v}.} \tag{148}$$

Combining (130) with this expression writes out all factors explicitly; no unknown configuration integral remains. The unconditional noised density is

$$P_t(x) = K_t(x) \int_{-1}^{1} p_0^q(q) e^{\ell_t \sqrt{N} q\mu} \mathcal{A}_{N-1}(\ell_t r R_q)\, \mathrm{d}q. \tag{149}$$

Both the HS posterior and the physical posterior therefore reduce exactly to one-dimensional integrals. This by itself does not say that their dependence on the observation is one-dimensional: the expressions contain both μ and r.

D.7 What symmetry closes exactly, and what it does not

Rotation invariance about u is preserved by the isotropic OU channel. Equation (149) exhibits it explicitly:

$$P_t(x) = G_t(\mu, r). \tag{150}$$

Here G_t is the value of the *ambient density* at a point with those coordinates. It is not the joint probability density of (μ, r). The latter includes the transverse volume factor and equals $\mathcal{S}_{N-2} r^{N-2} G_t(\mu, r)$ for $r > 0$. The distinction prevents us from inserting a radial Jacobian into the wrong score.

The chain rule gives, for $r > 0$,

$$F(x,t) = \nabla_x \log P_t(x) = (\partial_\mu \log G_t) u + (\partial_r \log G_t) \frac{x_\perp}{r}. \tag{151}$$

There are two scalar coefficient functions, but reconstructing the full vector still requires the transverse direction $x_\perp/r$. Thus this exact symmetry reduction is not a representation of the whole score vector using just two numbers and no geometric direction.

For the scalar projection, the exact OU identity is particularly simple:

$$\begin{aligned} u \cdot F(x,t) &= -\frac{u \cdot x}{\Delta_t} + \frac{e^{-t}}{\Delta_t} \mathbb{E}[u \cdot a \mid x] \\ &= -\frac{\mu}{\Delta_t} + \frac{\sqrt{N} e^{-t}}{\Delta_t} \mathbb{E}[q \mid x]. \end{aligned} \tag{152}$$

Therefore scalar closure of the projected drift only requires closure of $\mathbb{E}[q \mid x]$. It does *not* require every component of $\mathbb{E}[a \mid x]$ to be identical. For example, the exact conditional transverse mean is

$$\mathbb{E}[a_\perp \mid q, x] = R_q \frac{\mathcal{A}'_{N-1}(\ell_t r R_q)}{\mathcal{A}_{N-1}(\ell_t r R_q)} \frac{x_\perp}{r}, \qquad r > 0. \tag{153}$$

It follows by differentiating the angular log partition function in the observed transverse direction. The prime denotes derivative with respect to its scalar argument. At $r = 0$, symmetry gives zero transverse mean. This term is orthogonal to u, so it drops out of (152) exactly, without any approximation.

Exact cancellation for an idealized two-shell prior. If the clean physical magnetization were fixed exactly at $q = \pm m^*_{\mathrm{sph}}$, the two transverse radii would agree. The angular factors in (147) would therefore cancel:

$$\frac{P_t(x \mid q = +m^*_{\mathrm{sph}})}{P_t(x \mid q = -m^*_{\mathrm{sph}})} = \exp\left[2m^*_{\mathrm{sph}}\sqrt{N}\ell_t\mu\right]. \tag{154}$$

With prior weights $W_\pm$, this idealized model has exact branch and projected-score closure on μ at every $t > 0$. The actual finite-N Curie–Weiss prior is not this two-shell law: q fluctuates within each phase, so R_q fluctuates as well. We must control those fluctuations before using (154) for the original model. The next subsection supplies that control at the high-noise scale.

D.8 Controlled high-noise closure without conditional independence

We now keep $\beta > 1$, $h \in \mathbb{R}$, and $T > 0$ fixed, and study a sequence of forward times

$$t_N = \frac{1}{2}\log N + \tau_N, \qquad |\tau_N| \le L < \infty. \tag{155}$$

Here L is a fixed bound. For readability, write $t = t_N$ in this subsection. Then

$$e^{-t} = O(N^{-1/2}), \quad \ell_t = O(N^{-1/2}), \quad \chi_t := \sqrt{N}e^{-t} = O(1), \quad \Delta_t = T + O(N^{-1}).$$

All probabilistic error statements below are under the actual noised spherical law. The notation $O_{\mathbb{P}}(b_N)$ means that division by the positive scale b_N gives a sequence bounded in probability; $o_{\mathbb{P}}(1)$ means convergence to zero in probability. We first control the likelihood, then integrate the exact posterior to control the projected score. We do not infer derivative bounds merely from a small error in a log-density.

First control the observable transverse radius. Write the OU observation as $x = e^{-t}a + \sqrt{\Delta_t}\,\xi$, where $\xi \sim \mathcal{N}(0, I_N)$ is independent of a. With $\xi_u = u \cdot \xi$ and $\xi_\perp = \xi - \xi_u u$,

$$\begin{aligned} \mu &= \chi_t q + \sqrt{\Delta_t}\,\xi_u = O_{\mathbb{P}}(1), \\ r^2 &= e^{-2t}R_q^2 + 2e^{-t}\sqrt{\Delta_t}\,a_\perp \cdot \xi_\perp + \Delta_t\|\xi_\perp\|^2. \end{aligned} \tag{156}$$

The first term is $O(1)$ because $R_q^2 \le N$. Conditional on a, the middle term is centered Gaussian with variance $4e^{-2t}\Delta_t R_q^2 = O(1)$, hence is $O_{\mathbb{P}}(1)$. The last squared norm has a chi-square distribution with $N-1$ degrees of freedom, with mean $N-1$ and variance $2(N-1)$. It follows that

$$\boxed{\frac{r^2}{N-1} = \Delta_t + O_{\mathbb{P}}(N^{-1/2}).} \tag{157}$$

Thus r itself is large, of order $\sqrt{N}$, but its normalized squared value becomes nearly deterministic. This statement uses the OU noise and the fixed clean norm, not independent clean coordinates.

Then control the angular integral, rather than guessing a Gaussian. Let $d = N - 1$. For a uniformly distributed unit vector $\omega \in S^{d-1}$, symmetry makes its odd moments zero. The coordinate density from Appendix D.3, followed by the substitution $v = \omega_1^2$, gives

$$\begin{aligned} \mathbb{E}[\omega_1^{2k}] &= \frac{B(k+\frac{1}{2}, \frac{d-1}{2})}{B(\frac{1}{2}, \frac{d-1}{2})} \\ &= \frac{1 \cdot 3 \cdots (2k-1)}{d(d+2)\cdots(d+2k-2)}, \qquad k \ge 1. \end{aligned} \tag{158}$$

For $k = 0$ the moment and the empty products are one. Expanding the exponential in (133) and substituting these moments yields the convergent series

$$\mathcal{A}_d(z) = \sum_{k=0}^{\infty} \frac{(z^2/(2d))^k}{k!} \prod_{j=0}^{k-1} \frac{d}{d+2j}. \tag{159}$$

This permits an explicit uniform bound. Put $v = z^2/d \geq 0$. Since

$$0 \leq 1 - \prod_{j=0}^{k-1} \frac{d}{d+2j} \leq \sum_{j=0}^{k-1} \frac{2j}{d} = \frac{k(k-1)}{d},$$

comparison with the series for $e^{v/2}$ gives

$$0 \leq e^{v/2} - \mathcal{A}_d(z) \leq \frac{1}{d} \sum_{k=0}^{\infty} \frac{(v/2)^k}{k!} k(k-1) = \frac{v^2}{4d} e^{v/2}. \tag{160}$$

For v in any bounded interval, taking logarithms therefore gives

$$\log \mathcal{A}_d(z) = \frac{z^2}{2d} + O(d^{-1}), \qquad z^2/d = O(1). \tag{161}$$

The assumption is not $z \ll 1$. The argument z may grow like $\sqrt{d}$; it is z^2/d that must remain bounded. This matters because a high-dimensional angular integral is not a one-site small-field expansion.

For the spherical likelihood, $z = \ell_t r R_q$. On events with bounded r^2/N,

$$\frac{z^2}{N-1} = \frac{\ell_t^2 r^2 N(1-q^2)}{N-1} = O(1)$$

uniformly for $q \in [-1,1]$. Equations (161) and (157) therefore imply

$$\begin{aligned} \log \mathcal{A}_{N-1}(\ell_t r R_q) &= \frac{\ell_t^2 r^2 N(1-q^2)}{2(N-1)} + O_{\mathbb{P}}(N^{-1}) \\ &= \frac{N e^{-2t}}{2\Delta_t}(1-q^2) + O_{\mathbb{P}}(N^{-1/2}), \end{aligned} \tag{162}$$

with the last error uniform in q. Uniformity follows because $0 \leq 1 - q^2 \leq 1$ and $N\ell_t^2 = O(1)$.

The phase-dependent evidence is now scalar. Insert (162) into the exact likelihood (147). The q-independent part of the angular contribution cancels the term $-Ne^{-2t}/(2\Delta_t)$ in $\log K_t(x)$. Writing

$$\varphi_{\Delta_t}^{(N)}(x) := (2\pi\Delta_t)^{-N/2} e^{-\|x\|^2/(2\Delta_t)},$$

we obtain

$$\boxed{\begin{aligned} \log P_t(x \mid q) = \log \varphi_{\Delta_t}^{(N)}(x) + \frac{\chi_t}{\Delta_t} q\mu - \frac{\chi_t^2}{2\Delta_t} q^2 + \epsilon_N(q,x), \\ \sup_{q \in [-1,1]} |\epsilon_N(q, X_t)| = O_{\mathbb{P}}(N^{-1/2}). \end{aligned}} \tag{163}$$

Only the term linear in q contains a leading random observation statistic, namely μ. The quadratic q^2 correction is deterministic at fixed time. This is the same evidence structure obtained for the three-state branches, but it has been derived here without a product law for the clean spins.

Use the posterior integral to obtain a controlled mean. After removing the common Gaussian factor, Bayes' rule becomes

$$p_t^q(q \mid x) = \frac{p_0^q(q) e^{L_t(q,\mu) + \epsilon_N(q,x)}}{\displaystyle\int_{-1}^{1} p_0^q(v) e^{L_t(v,\mu) + \epsilon_N(v,x)} \, \mathrm{d}v}, \qquad L_t(q,\mu) := \frac{\chi_t}{\Delta_t} q\mu - \frac{\chi_t^2}{2\Delta_t} q^2. \tag{164}$$

On any event with bounded $|\mu|$, the function $e^{L_t(q,\mu)}$ is bounded above and away from zero, and is Lipschitz in q, uniformly on $[-1,1]$. The uniform error in (163) changes normalized expectations of bounded functions by $o_{\mathbb{P}}(1)$. The clean two-phase concentration in (140) then permits replacement of the q integrals by their values at $q = \pm m_{\mathrm{sph}}^*$. Because $\mu = O_{\mathbb{P}}(1)$, restricting first to bounded $|\mu|$ loses arbitrarily little probability. This explains why Laplace concentration can be used after the noisy reweighting, rather than simply assuming the clean delta approximation remains valid.

Let $\varepsilon = \pm 1$ denote the two phase labels. Their leading posterior weights are

$$\pi_\varepsilon(\mu,t) = \frac{W_\varepsilon e^{\varepsilon\chi_t m^*_{\rm sph}\mu/\Delta_t - (\chi_t m^*_{\rm sph})^2/(2\Delta_t)}}{\sum_{\delta=\pm1} W_\delta e^{\delta\chi_t m^*_{\rm sph}\mu/\Delta_t - (\chi_t m^*_{\rm sph})^2/(2\Delta_t)}}. \tag{165}$$

The squared-mean corrections are equal and cancel. Using $W_\varepsilon \propto e^{\varepsilon h m^*_{\rm sph}}$, define

$$A_{\rm sph}(\mu,t) := m^*_{\rm sph}\left(h + \frac{\chi_t}{\Delta_t}\mu\right).$$

Then $\pi_\varepsilon = e^{\varepsilon A_{\rm sph}}/(2\cosh A_{\rm sph})$. The identity $(e^A - e^{-A})/(e^A + e^{-A}) = \tanh A$ gives

$$\boxed{\mathbb{E}[q \mid X_t] = m^*_{\rm sph}\tanh A_{\rm sph}(\mu,t) + o_{\mathbb{P}}(1).} \tag{166}$$

Substitution into the *exact* projected-score identity (152) proves

$$\boxed{u\cdot F(X_t,t) = -\frac{\mu}{\Delta_t} + \frac{\chi_t m^*_{\rm sph}}{\Delta_t}\tanh A_{\rm sph}(\mu,t) + o_{\mathbb{P}}(1).} \tag{167}$$

The prefactor χ_t/Δ_t is bounded in the chosen window, so it does not amplify the posterior-mean error. This is a statement about the projection of the full microscopic score at typical configurations. It is stronger than deriving a marginal score, but weaker than asserting an exact scalar representation of the whole vector field.

Precisely what has been controlled

The result is a finite-time, large-N projected-drift approximation for $t_N = \frac{1}{2}\log N + \tau_N$ with bounded τ_N, fixed $\beta > 1$, and fixed h, T. It is not a uniform approximation for every $x \in \mathbb{R}^N$, not a result in the critical window $\beta \to 1$, and not a statement valid down to $t = 0$. Convergence of complete reverse trajectories on a time interval would require additional temporal control. No such pathwise convergence claim is needed to derive the leading local force below.

D.9 Project the reverse drift and integrate the effective potential

We use the increasing reverse clock $s = t_f - t$ and the full reverse drift $B(x,t) = x + 2TF(x,t)$ from Appendix A.3. Its projection is exactly

$$u\cdot B(x,t) = \mu + 2T\,u\cdot F(x,t).$$

Using (167), define the leading scalar force

$$\begin{aligned}\mathcal{B}_{\rm sph}(\mu,t) &:= \left(1 - \frac{2T}{\Delta_t}\right)\mu + \frac{2T\chi_t m^*_{\rm sph}}{\Delta_t}\tanh A_{\rm sph}(\mu,t),\\ u\cdot B(X_t,t) &= \mathcal{B}_{\rm sph}(\mu,t) + o_{\mathbb{P}}(1).\end{aligned} \tag{168}$$

Because u has norm one, $u\cdot\overline{W}_s$ is a standard one-dimensional Brownian motion W^μ_s:

$$\mathrm{d}\langle u\cdot\overline{W}\rangle_s = \sum_i u_i^2\,\mathrm{d}s = \mathrm{d}s.$$

The reduced leading equation is therefore

$$\mathrm{d}\mu_s = \mathcal{B}_{\rm sph}(\mu_s, t_f - s)\,\mathrm{d}s + \sqrt{2T}\,\mathrm{d}W^\mu_s. \tag{169}$$

This is the reduced model obtained from the drift approximation, not an assertion that the finite-N microscopic drift has become exactly scalar. The ordinary mean $M(x)$ and $\mu = \sqrt{N}M(x)$ contain the same information; the normalization makes the projected Brownian noise order one.

At fixed t, seek $V_{\mathrm{sph},\Delta}$ with $-\partial_\mu V_{\mathrm{sph},\Delta} = \mathcal{B}_{\mathrm{sph}}$. The linear term integrates to $\frac{1}{2}(2T/\Delta_t - 1)\mu^2$. For the nonlinear term,

$$\partial_\mu A_{\mathrm{sph}} = \frac{m^*_{\mathrm{sph}}\chi_t}{\Delta_t}, \qquad \frac{\mathrm{d}}{\mathrm{d}A}\log\cosh A = \tanh A.$$

The chain rule gives

$$\partial_\mu[-2T\log\cosh A_{\mathrm{sph}}] = -\frac{2Tm^*_{\mathrm{sph}}\chi_t}{\Delta_t}\tanh A_{\mathrm{sph}}.$$

Thus the potential of the leading reduced model is

$$\boxed{\begin{aligned} V_{\mathrm{sph},\Delta}(\mu,t) = \frac{1}{2}\left(\frac{2T}{\Delta_t} - 1\right)\mu^2 \\ - 2T\log\cosh\left[m^*_{\mathrm{sph}}\left(h + \frac{\sqrt{N}e^{-t}}{\Delta_t}\mu\right)\right] + C(t). \end{aligned}} \tag{170}$$

The function $C(t)$ is arbitrary because its μ derivative vanishes. Differentiating (170) reproduces both terms of (168) with their signs and factors of $2T$. Keeping Δ_t here does not remove the preceding high-noise and thermodynamic approximations; it merely avoids one additional replacement.

Since $\Delta_t = T + O(N^{-1})$ in the chosen time window, the simpler leading potential is

$$\boxed{V_{\mathrm{sph}}(\mu,t) = \frac{1}{2}\mu^2 - 2T\log\cosh\left[m^*_{\mathrm{sph}}\left(h + \frac{\sqrt{N}e^{-t}}{T}\mu\right)\right] + C(t).} \tag{171}$$

Its form is identical to the Ising expression. The different microscopic entropy enters through $m^*_{\mathrm{sph}} = \sqrt{1 - 1/\beta}$ rather than through the Ising equation $m^* = \tanh(\beta m^*)$. The hyperbolic tangent in the reduced force now comes from posterior competition between two branches, not from a binary single-site spin sum.

Stability means curvature of the force potential, not its value. At $h = 0$, the origin is stationary in the leading reduced equation. For the simplified potential,

$$\partial^2_\mu V_{\mathrm{sph}}(0,t) = 1 - \frac{2N(m^*_{\mathrm{sph}})^2}{T}e^{-2t}. \tag{172}$$

A small deterministic perturbation δ_s around the origin obeys

$$\frac{\mathrm{d}\delta_s}{\mathrm{d}s} = -\partial^2_\mu V_{\mathrm{sph}}(0, t_f - s)\delta_s, \qquad \delta_s = \delta_0\exp\left[-\int_0^s \partial^2_\mu V_{\mathrm{sph}}(0, t_f - v)\,\mathrm{d}v\right].$$

Positive instantaneous curvature damps perturbations; negative curvature amplifies them. The instantaneous curvature threshold in this leading model is

$$t^{\mathrm{sph}}_{\mathrm{bif}} = \frac{1}{2}\log\left(\frac{2N(m^*_{\mathrm{sph}})^2}{T}\right) = \frac{1}{2}\log N + O(1). \tag{173}$$

This is a local deterministic bifurcation scale, not an exact stochastic selection time or a prescribed recovery-confidence threshold. For $h \neq 0$ the origin is generally not stationary and this symmetric curvature test cannot be applied unchanged.

D.10 Exact projected marginal: independent check and the two meanings of closure

There is an exact scalar process that exists independently of the preceding high-noise argument. Project the *forward* OU process:

$$\mu_t = \chi_t q(a) + \sqrt{\Delta_t}\,\xi_u, \qquad \mu_t \mid q \sim \mathcal{N}(\chi_t q, \Delta_t).$$

Let $\varphi_\Delta(z) := (2\pi\Delta)^{-1/2} e^{-z^2/(2\Delta)}$. The exact noisy magnetization density is

$$\boxed{\begin{aligned} p_t^\mu(\mu) &= \int_{-1}^{1} p_0^q(q)\varphi_{\Delta_t}(\mu - \chi_t q)\,\mathrm{d}q \\ &= \frac{C_N}{Z_N^{\mathrm{sph}}\sqrt{2\pi\Delta_t}} \int_{-1}^{1} (1-q^2)^{(N-3)/2} \\ &\qquad \times e^{\beta N q^2/2 + hq - (\mu - \chi_t q)^2/(2\Delta_t)}\,\mathrm{d}q. \end{aligned}} \tag{174}$$

Since μ_t is itself an exact forward OU process, its reverse drift relative to its own state information is $\mu + 2T\partial_\mu \log p_t^\mu(\mu)$. Its exact scalar potential is

$$V_{\mathrm{sph,marg}}(\mu, t) = -\frac{1}{2}\mu^2 - 2T \log p_t^\mu(\mu) + C(t), \qquad t > 0. \tag{175}$$

This statement holds at finite N without replacing the magnetization by two branches. It is a potential for the marginal reverse process, not an equality of the pointwise full-state projected drift with a scalar function.

The connection can be made explicit. Integrating over the perpendicular coordinates and differentiating the smooth OU density gives

$$\partial_\mu \log p_t^\mu(\mu) = \mathbb{E}[u \cdot F(X_t, t) \mid u \cdot X_t = \mu]. \tag{176}$$

Thus the exact scalar marginal score is a *conditional average* of the projected microscopic score. The high-noise result (167) is stronger: the projected microscopic score itself becomes a function of μ, up to a vanishing error for typical X_t. Conditioning on the full microscopic state and conditioning only on its projection need not give the same finite-N drift.

As an independent check, use phase concentration inside (174) at the high-noise scale:

$$p_t^\mu(\mu) \simeq W_+ \varphi_{\Delta_t}(\mu - c_t) + W_- \varphi_{\Delta_t}(\mu + c_t), \qquad c_t := \chi_t m_{\mathrm{sph}}^*.$$

The within-phase fluctuations of q have width $N^{-1/2}$ and are multiplied by $\chi_t = O(1)$, so their contribution to μ vanishes. Complete the squares in the two Gaussian factors:

$$-\frac{(\mu \mp c_t)^2}{2\Delta_t} = -\frac{\mu^2}{2\Delta_t} \pm \frac{c_t \mu}{\Delta_t} - \frac{c_t^2}{2\Delta_t}.$$

Using the weights in (140),

$$p_t^\mu(\mu) \simeq \frac{e^{-(\mu^2 + c_t^2)/(2\Delta_t)}}{\sqrt{2\pi\Delta_t}} \frac{\cosh(h m_{\mathrm{sph}}^* + c_t\mu/\Delta_t)}{\cosh(h m_{\mathrm{sph}}^*)}.$$

Substituting this expression into (175), and absorbing only terms independent of μ into $C(t)$, reproduces (170) exactly for this approximate Gaussian-mixture density. The two derivations therefore agree at their common approximation order.

D.11 What the spherical comparison establishes

The original Ising derivation and the spherical extension change the microscopic reference measure while retaining the same mean-field interaction. Their relation can be summarized as follows.

Step	Ising	Spherical
Clean constraint	$a_i \in \{-1, +1\}$	$\|a\|_2^2 = N$
Reference measure	Discrete counting measure	Normalized surface measure
Interaction	$\beta S^2/(2N) + (h/N)S$	The same interaction
HS identity	Exact scalar Gaussian integral	The same exact identity
Given HS variable	Coordinates independent	Norm constraint retains dependence
Auxiliary marginal	Gaussian factor times a power of $2\cosh$	Gaussian factor times $\mathcal{J}_N$
Physical magnetization	Binomial multiplicity	Surface factor $(1-q^2)^{(N-3)/2}$
Phase location, $\beta > 1$	$m_{\mathrm{I}}^* = \tanh(\beta m_{\mathrm{I}}^*)$	$m_{\mathrm{sph}}^* = \sqrt{1 - 1/\beta}$
Diffused evidence	Product of one-site factors	Transverse angular integral
Route to scalar closure	High-noise branch log-odds	Angular cancellation and concentration
Leading potential	Quadratic minus $2T\log\cosh$	Same form, different m^*

The facts established here are exact reference-measure and HS formulas, exact latent likelihoods and posteriors, and an exact two-statistic symmetry description of the ambient noisy density. The one-statistic projection closure is then a controlled high-noise result for fixed $\beta > 1$. We have not shown that the entire score vector can be reconstructed from μ alone, nor that the scalar approximation remains valid at arbitrary noise levels.

The mechanism that survives the change of geometry

The rank-one interaction selects the uniform direction u. The clean magnetization concentrates into two symmetry-related phases. After OU noise, the transverse angular contribution either cancels exactly for fixed opposite magnetizations or becomes a controlled deterministic correction in the actual high-dimensional model. Consequently the posterior mean of the physical magnetization closes on μ to leading order. The exact projection identity then closes the reverse force, and ordinary one-variable integration gives its potential. Conditional independence is one route to this mechanism, but the spherical calculation shows that it is not required for this particular high-noise projected closure.

The number of phases and the microscopic geometry are therefore separate axes of generalization. The three-state calculation tests the former; this spherical calculation tests the latter. A broader n-phase result would still need its own assumptions on phase concentration, observation statistics, and the intended projected force. No conclusion about arbitrary geometries or all diffusion times follows solely from the common $\log\cosh$ expression.